

Implied Volatility Changes and Corporate Bond Returns

Jie Cao,^a Amit Goyal,^{b,*} Xiao Xiao,^c Xintong Zhan^d

^aFaculty of Business, School of Accounting and Finance, Hong Kong Polytechnic University, Kowloon, Hong Kong; ^bUniversity of Lausanne and Swiss Finance Institute, 1015 Lausanne, Switzerland; ^cBayes Business School, City University London, London EC1Y 8TZ, United Kingdom; ^dSchool of Management, Fudan University, Shanghai 200433, China

*Corresponding author

Contact: jie.cao@polyu.edu.hk,  <https://orcid.org/0000-0002-5245-8895> (JC); amit.goyal@unil.ch,  <https://orcid.org/0000-0002-4004-8039> (AG); shawny.xxiao@gmail.com,  <https://orcid.org/0000-0002-0564-9795> (XX); xintongzhan@fudan.edu.cn,  <https://orcid.org/0000-0003-2787-4464> (XZ)

Received: August 26, 2020

Revised: March 21, 2021

Accepted: May 25, 2021

Published Online in Articles in Advance:
March 25, 2022

<https://doi.org/10.1287/mnsc.2022.4379>

Copyright: © 2022 INFORMS

Abstract. Corporate bonds with large increases in implied volatility over the past month underperform those with large decreases in implied volatility by 0.6% per month. In contrast to existing studies that show implied volatility changes carry information about fundamental news, our evidence suggests that implied volatility changes contain information about uncertainty shocks to the firm. Our results are consistent with the notion that informed traders with new information about firm risk prefer to trade in the option market and the corporate bond market underreacts to this information.

History: Accepted by Haoxiang Zhu, finance.

Funding: We thank the Canadian Derivatives Institute for financial support. J. Cao and X. Zhan acknowledge generous financial support of the Research Grant Council of the Hong Kong Special Administrative Region, China [Grants GRF 14501720, 14500919].

Supplemental Material: The online appendix and data are available at <https://doi.org/10.1287/mnsc.2022.4379>.

Keywords: corporate bonds • implied volatility changes • default risk • information diffusion

1. Introduction

Options are redundant assets only in perfect markets (Black and Scholes 1973, Merton 1973). Real world frictions, such as transaction costs, short-sale constraints, and segmented markets, may make informed traders migrate to options markets instead of stock or bond markets (Back 1993, Figlewski and Webb 1993, Biais and Hillion 1994). Numerous studies examine the information transmission from the option market to stock prices and show predictability of future stock returns from various option-related variables.¹ However, whether an option market has relevant information for the future corporate bond return and, hence, contributes to the price discovery of corporate bonds has received much less attention. The size of the corporate bond market is nontrivial. The outstanding amount of corporate bonds issued by nonfinancial corporations was \$5.8 trillion at the end of 2019 (see table L.213 in the financial accounts of the United States, Federal Reserve Board Z.1 flow of funds). In this paper, we document that option prices contain important information about the default risk of the underlying firm. The underlying firms' bond prices underreact to such information. Overall, we show that changes in the implied volatility of equity options predict underlying firms' bond return even after adjusting for the bond risk factors proposed by Bai et al. (2019).

The predictability from option-implied volatility to stock or bond returns is consistent with economies, such as in Easley et al. (1998), in which informed traders choose to trade in the option market before other markets. Change in implied volatility could carry information about fundamental news or changing risk of the underlying firm. Consistent with the fundamental news story, An et al. (2014) show that change in implied volatilities of call ($\Delta CVOL$) or put ($\Delta PVOL$) options predict high or low future underlying stock returns, respectively. We, instead, posit that changes in implied volatility convey information also about the changes in default risk of the underlying firm.² In other words, an increase in $CVOL$ could be either due to good fundamental news or about increasing risk, whereas an increase in $PVOL$ could be either due to bad fundamental news or about increasing risk. The common component of $\Delta CVOL$ and $\Delta PVOL$, measured as $(\Delta CVOL + \Delta PVOL)/2$ (we relabel it as $\Delta ImpVOL$), thus, is likely to signal increasing risk of the underlying firm. In our sample, the cross-sectional correlation between $\Delta CVOL - \Delta PVOL$ and $\Delta CVOL + \Delta PVOL$ is almost zero. We find that $\Delta ImpVOL$ predicts low future bond returns, but $\Delta CVOL - \Delta PVOL$, the main variable in An et al. (2014), does not predict future bond returns. (We also find $\Delta ImpVOL$ does not predict future stock returns.) Thus, different implied volatility changes carry different information for stock and bond returns.

Beyond the different information content embedded in changes in implied volatilities, there are at least three reasons why predictability might be different in stock and bond markets. First, stock and bond returns are not perfectly correlated (the average cross-sectional correlation in our sample is only 0.44). Corporate bonds with less credit risk have smaller hedge ratios and comove less with the stock. Second, the corporate bond market also consists of more institutional and sophisticated investors than the stock market. Edwards et al. (2007) document a median trade size of \$240,600 in the corporate bond market and find that, in the corporate bond market, transaction costs are lower for larger trades, suggesting that institutions are likely to be the typical traders in bonds. Third, following the seminal insights from Black and Scholes (1973) and Merton (1973), it is important to remember that a stock is a long position of call option on the firm's asset, whereas a bond is a short position of put option on the firm's asset. Other things equal, volatility of firm value is detrimental to bondholders as it increases the chance of default, but volatility has a positive impact on stockholders (Campbell and Taksler 2003). As such, changes in implied volatilities could have different impacts on bond and stock prices.³ Therefore, we believe that it is an open question whether the option market leads the corporate bond market and the nature of information transmitted if any.

We study the predictability of the sample of corporate bonds from the Trade Reporting and Compliance Engine (TRACE) over the period 2002 to 2017. At the end of each month, we sort all bonds into decile portfolios based on $\Delta ImpVOL$ over the previous month. We keep these portfolios for one month and rebalance each month. We find that the bonds in the top decile (largest increase in implied volatility) underperform those in the bottom decile (largest decrease in implied volatility) by 0.60% (t -statistic = 3.62). To ensure that the return differences are not a compensation for risk, we use a bond and stock market factor model. The stock market factors are the six factors from Fama and French (2018). The bond market factors include bond market return, downside risk, credit risk, liquidity risk, and reversal factor from Bai et al. (2019). After controlling for these 11 stock and bond market factors, the risk-adjusted return spread between the top and bottom decile is even more significant and is at 0.98% (t -statistic = 5.57) per month.

Prior literature shows that bond characteristics such as maturity, coupon, age, and ratings can explain the cross-section of corporate bond returns (see, for example, Gebhardt et al. 2005a, b). Therefore, we test whether the negative relation between $\Delta ImpVOL$ and future bond returns still holds after controlling for bond characteristics used in Bai et al. (2019). Bivariate

portfolio sorts indicate that implied volatility change remains a significant predictor of future bond returns after controlling for bond characteristics, such as size, maturity, credit rating, liquidity, and lagged bond return. Although we do not observe strong patterns in the portfolios sorted by illiquidity or lagged bond return in the past month, we do find that the absolute return spread is larger for bonds with longer maturity and higher credit risk. The finding that the predictability is higher for lower rated bonds is similar to that in Chordia et al. (2017) and Jostova et al. (2013).

In recent work, Chung et al. (2019) find that bonds with high volatility betas or low idiosyncratic bond volatility have higher expected returns. To allay the concern that correlations between different volatility variables drive our results, we also control for bond volatility, bond idiosyncratic volatility, stock implied volatility, stock idiosyncratic volatility, and VIX beta. We do find that the absolute return spread sorted on the implied volatility changes is related to many of these other volatility variables. However, our results on $\Delta ImpVOL$ cannot be fully explained by these alternative volatility proxies because the effect of changes in implied volatilities is robust to controlling for these volatility variables. We further confirm the results from portfolio sorts using regression approach. We control for multiple variables simultaneously in the Fama and MacBeth (1973) regressions. The predictability of $\Delta ImpVOL$ is robust after controlling for all these bond characteristics and all bond and volatility characteristics.

Next, we examine why $\Delta ImpVOL$ negatively predicts future bond returns. We find that options markets have relevant information to predict default risk, which is particularly important for determining bond returns. In particular, we find that $\Delta ImpVOL$ significantly predicts changes of probability of default and changes in rating downgrades in the next month. Our findings are consistent with the literature that uses equity volatility as one of the inputs in corporate default prediction (see, for example, Shumway 2001, Campbell et al. 2008). Our results also echo those of Campbell and Taksler (2003) who show that equity volatility explains as much of the level of bond yields as do credit ratings. In contrast to this line of enquiry, we use forward-looking volatility estimates to predict changes in bond prices. Our evidence suggests that the option market leads the price discovery process of the corporate bond market because it reflects the changes of default risk faster than the corporate bond market. We also find that implied volatility changes predict changes in credit default swap (CDS) spreads. Given that CDS are highly sensitive to default risk changes, this predictability further supports the hypothesis that increase in implied volatility predicts higher default risk of the underlying firm, leading to

lower bond return and an increase in CDS price (spread).

A natural question is why corporate bonds fail to impound the relevant information about the change in default risk into bond prices. The slow adjustment of bond prices that we document might reflect slow diffusion of information from options to bonds or limits to arbitrage in the bond market. Whereas we are not able to conclusively disentangle these two hypotheses, we find that both explanations play a role in explaining the predictability.

The hypothesis of slow diffusion of information is supported by four pieces of evidence. First, consistent with the implication of the sequential trading model in Easley et al. (1998), we find that the bond return predictability is the highest when option trading volumes increase the most and when bond trading volumes decrease the most. This suggests that some informed investors choose to trade options before trading in the bond market. Second, whereas bond return predictability is stronger among less liquid bonds, we find predictability in even very liquid bonds further pointing to slow diffusion of information as a likely cause of predictability. Third, we find that predictability is five to eight times stronger on rating announcement days than on other days. These findings are consistent with the idea that biased expectations drive our bond portfolio returns, and they are partially corrected upon salient news arrival (Engelberg et al. 2018). Fourth, we find that firms with high investor attention, measured by dual institutional ownership of both stock and bond, exhibit weaker bond return predictability, suggesting that investors' inattention can partially explain the predictability of change in implied volatility.

Transaction cost analysis provides a rationale for why arbitrageurs do not enforce price efficiency in the bond market. Whereas returns net of the transaction costs of Edwards et al. (2007) are still positive and significant for large trade size (\$1M), net returns to the trading strategy are not positive for smaller trade sizes (\$100K) or for the transaction cost estimates of Bao et al. (2011). Thus, slow diffusion of information because of investor inattention coupled with high limits to arbitrage explains why bond prices do not incorporate the information in the change in default risk embedded in option prices.

Our paper makes three contributions. First, we contribute to the literature that uses equity volatility in corporate default prediction models (see, for example, Shumway 2001, Chava and Jarrow 2004, Campbell et al. 2008). Campbell and Taksler (2003) and Zhang et al. (2009) show that equity volatility is useful for explaining cross-sectional variation in bond yields and spreads, and Wang et al. (2013) show that variance risk premium contributes to determine the credit spreads. Relatedly, Cremers et al. (2008) find that options contain useful

information for predicting credit spreads although these authors focus on jump risk rather than volatility changes. Whereas these studies are primarily concerned with explaining the level of bond prices (or yield spreads), they do open the possibility that (forecasts of) changes in volatility can predict changes in bond prices/yields (returns). Our paper takes this next step. Implied volatility from options is well-known to be a good predictor of future volatility (Christensen and Prabhala 1998). We believe that ours is the first study to show that option-implied volatility contains useful information for bond returns for a comprehensive sample of U.S. corporate bonds.⁴ Importantly, in contrast to the extant literature, we also explore the entire price discovery process of how corporate bonds incorporate option information by showing the direction of information flow, the impact of investor inattention, and the role of limits to arbitrage in information transmission.

Second, we document a new predictor for the cross-section of corporate bond returns. Gebhardt et al. (2005a) find that some bond characteristics predict bond returns in addition to risk-related variables. Studies analyzing predictability of the cross-section of bond returns and prices from bond and stock characteristics include Bali et al. (2021), Choi and Kim (2018), Chordia et al. (2017), Chung et al. (2019), Jostova et al. (2013), and Kwan (1996).⁵ Bai et al. (2019) instead focus on risk-based explanations in proposing a new bond factor pricing model. Our paper differs from these studies in that we examine the information content from the derivatives market, which is generally regarded as more informative. Our paper also complements recent studies on factor investing in corporate bonds by constructing bond factors from options.⁶

Third, we contribute to the literature that studies information transmission of options to the stock market. For example, several studies show that option-related variables can predict stock returns, such as the volatility spread in Bali and Hovakimian (2009), deviation of put-call parity in Cremers and Weinbaum (2010), volatility smirk in Xing et al. (2010), option-to-stock volume ratio in Johnson and So (2012), and stock order imbalance induced by option transactions in Hu (2014). We complement these studies by examining the information content from implied volatility changes on the corporate bond returns. As mentioned earlier, our paper is closely related to An et al. (2014), who also study the predictive content of change in implied volatilities. Beyond the obvious difference that An et al. (2014) focus on stock returns and we study bond returns, we also show that the information content in implied volatilities used in the two papers is different. In independent work, Navon (2014) also studies the relation between implied volatility changes and corporate bond returns. We study a much bigger cross-section (20 times Navon's (2014) sample) over a

longer sample period. More importantly, in contrast to Navon (2014), we also explore the economic channels of predictability. For example, we relate our results to those of An et al. (2014) and show that $\Delta CVOL + \Delta PVOL$ contains default-related information, whereas $\Delta CVOL - \Delta PVOL$ is related to fundamental news, and we test the default risk channel by testing the predictability of $\Delta CVOL + \Delta PVOL$ for changes in *EDF*, *CDS* spreads, etc.

The remainder of this paper is organized as follows. Section 2 describes our data and the construction of variables. Section 3 discusses the empirical evidence on the predictability of changes in implied volatilities for corporate bond returns. We discuss information content of the change in implied volatility in Section 4. Section 5 tests the slow diffusion of information and limits to arbitrage explanations for why the bond prices underreact. Section 6 concludes.

2. Data and Variables

2.1. Corporate Bond Data and Bond Return

We obtain corporate bond data from the enhanced version of TRACE for the sample period July 2002 to August 2017. Enhanced TRACE offers more trade records that span the entire over-the-counter market from earlier than those in the standard TRACE. Moreover, trade volumes are reported accurately and are not capped at certain levels based on bond ratings in enhanced TRACE (Bessembinder et al. 2006). We merge the enhanced TRACE data set with the Mergent Fixed Income Securities Database (FISD) with issuance information for all fixed-income securities that have a committee on uniform security identification procedures number. The FISD data set contains bond characteristics, such as offering amount, offering date, maturity date, coupon rate, coupon type, interest payment frequency, bond type, and bond rating.

We follow Bai et al. (2019) and remove the following observations: (i) bonds that are not listed or traded in the U.S. public market; (ii) bonds that are structured notes, mortgage-backed, asset-backed, agency-backed, or equity-linked; (iii) convertible, sinking fund bonds, and bonds with a floater or odd frequency of coupon payments; (iv) bonds that trade under \$5 or above \$1,000; (v) bonds that have less than one year to maturity; (vi) intraday bond transactions that are labeled as when issued, locked in, or have special sales conditions, that have more than a two-day settlement, that are canceled and adjust records that are subsequently corrected or reversed.⁷ Similar to Chordia et al. (2017), we study the cross-section of bond returns instead of event analysis and, therefore, do not impose filters to remove bond transaction with trading volume smaller than \$10,000 as in Bessembinder et al. (2006).⁸

Following Bessembinder et al. (2009), we calculate the trading volume-weighted average of intraday bond prices as daily prices. This approach puts more weights on the trades with low transaction costs and should more accurately reflect the bond price. To calculate monthly bond return, we use only the last observation during the last five trading days of each month. If there is no observation during these five days, the bond price is set to be missing.

The monthly corporate bond return at month t is calculated as

$$r_t = \frac{P_t + AI_t + C_t}{P_{t-1} + AI_{t-1}} - 1, \quad (1)$$

where P_t is the transaction price of the bond at the end of month t , AI_t is the accrued interest, and C_t is the coupon payment from the end of month $t-1$ to the end of month t . We denote R_t as bond excess return, $R_t = r_t - r_{ft}$, where r_{ft} is the risk-free rate proxied by the one-month treasury bill rate.

Besides bond return, we construct several bond characteristics using the TRACE and the FISD data. We calculate *Size* as the logarithm of offering amount of the bond. *Rating* is calculated as the numerical rating score provided by Moody's and Standard and Poor's (S&P).⁹ A numerical score of one refers to an AAA rating by S&P and Aaa rating by Moody. A numerical score of 21 refers to a C rating for both S&P and Moody. Ratings of 10 or below are considered as investment grade, and ratings above 10 are considered as non-investment grade. *Maturity* is the time to maturity of the bond in years. *Illiquidity* is the autocovariance of the daily log bond price change in each month multiplied by -1 as defined in Bao et al. (2011). *Lag return* is the bond return in the past month. Finally, *VaR* (5%) is the 5% value-at-risk of corporate bond return, defined as the second lowest monthly return over the past 36 months as in Bai et al. (2019).

2.2. Option Data

We obtain daily implied volatility data from the volatility surface in OptionMetrics. OptionMetrics provides the interpolated volatility surface for each stock on each day, using a kernel smoothing algorithm and options with various strike prices and maturities. Implied volatilities are calculated based on the industry-standard Cox–Ross–Rubinstein binomial tree model. This model can accommodate underlying securities with either discrete dividend payments or a continuous dividend yield and American style stock options with early exercise features. This volatility surface data set contains information of volatilities with various maturities and deltas. An implied volatility is only included if there are enough option price data on that date to accurately interpolate the required values. One advantage of using the volatility surface

data is that the maturities and deltas are fixed for each trading day, and hence, there is no need to control for variations in expiration dates and strike prices.

In the main analysis of this study, we use implied volatilities with a delta of 0.5 and an expiration of 365 days for put and call options. We use month-end observation to calculate the changes in implied volatilities, which we denote as $\Delta CVOL$ for call options and $\Delta PVOL$ for put options, respectively. To avoid price pressure bias (Goncalves-Pinto et al. 2020), we use implied volatility one day before the date of the observation used to calculate bond price. In other words, we skip a day between option information and portfolio formation. Skipping two days makes no material difference to our results.¹⁰

After merging the corporate bond data from TRACE with the option data from Option-Metrics, we have 881,625 bond-month observations from July

2002 to August 2017. The number of unique firms in our sample is 2,327. In Panel A of Table 1, we report the summary statistics of bond returns, changes in implied volatilities, and various bond characteristics. Bonds in the sample have an average return of 0.51%, an average rating of 8.52 (BBB+), and an average maturity of 8.54 years. The average of implied volatility of put and call options is 0.31. The changes in implied volatilities of call and put options are both, on average, close to zero. The summary statistics are similar to those in the prior literature. Recall that we do not include bonds with less than one year to maturity in our sample. At the same time, maturity information is missing for some bonds as evidenced by fewer observations for the *Maturity* variable. Including only the bonds for which we have the maturity has an insignificant impact on our results as discussed in next sections.

Table 1. Descriptive Statistics

Panel A: Summary statistics											
Variable	N	Mean	Standard deviation	10th percentile	Lower quartile	Median	Upper quartile	90th percentile			
Return	881,625	0.51	3.20	−2.04	−0.45	0.34	1.48	3.25			
ΔCVOL	881,625	−0.08	5.02	−3.67	−1.62	−0.18	1.33	3.57			
ΔPVOL	881,625	−0.10	5.20	−3.79	−1.69	−0.25	1.24	3.64			
ΔCVOL − ΔPVOL	881,625	0.01	4.26	−1.80	−0.55	0.06	0.68	1.87			
ΔImpVOL	881,625	−0.09	3.89	−3.43	−1.57	−0.21	1.17	3.39			
ImpVOL	881,625	30.70	14.67	17.65	21.16	26.56	35.68	47.80			
Size	832,812	12.51	1.62	9.85	12.21	12.90	13.53	14.04			
Rating	741,028	8.52	4.45	4.00	6.00	8.00	10.00	14.00			
Maturity	832,778	8.54	8.63	1.17	2.76	5.63	9.68	23.69			
Illiquidity	801,804	0.95	2.73	0.00	0.02	0.16	0.72	2.27			
Lag return	845,063	0.49	3.12	−2.00	−0.44	0.34	1.44	3.17			
VaR (5%)	694,870	−4.01	4.28	−8.44	−4.81	−2.76	−1.49	−0.78			
Panel B: Time-series average of cross-sectional correlations											
	ΔCVOL	ΔPVOL	ΔCVOL−ΔPVOL	ΔImpVOL	ImpVOL	Size	Rating	Maturity	Illiquidity	Lag return	VaR (5%)
Return	−0.04	−0.04	0.00	−0.04	0.04	−0.02	0.06	0.04	0.03	−0.06	−0.08
ΔCVOL		0.58	0.42	0.87	0.08	0.00	−0.02	0.00	0.01	−0.13	0.02
ΔPVOL			−0.41	0.88	0.11	0.00	−0.01	0.01	0.01	−0.14	0.01
ΔCVOL − ΔPVOL				0.00	−0.02	−0.00	−0.01	0.00	0.00	0.01	0.02
ΔImpVOL					0.11	0.00	−0.02	0.01	0.01	−0.15	0.01
ImpVOL						−0.03	0.50	−0.09	0.07	0.00	−0.47
Size							0.02	−0.03	−0.23	−0.02	0.14
Rating								−0.10	0.04	0.05	−0.38
Maturity									0.19	0.03	−0.29
Illiquidity										0.01	−0.22
Lag Return											−0.03

Notes. This table reports the summary statistics of bond return, changes in implied volatilities, and bond characteristics. *Return* is monthly bond return reported in percentage per month. $\Delta CVOL$ ($\Delta PVOL$) is the change of implied volatility of at-the-money call (put) option with 365 days of maturity; the change is calculated using the month-end observations. *ImpVOL* ($\Delta ImpVOL$) is the (change of) average implied volatility of at-the-money call and put options with 365 days of maturity. *Size* is the logarithm of the offering amount of the bond. *Rating* is the numerical rating score, for which 1 refers to an AAA rating by S&P and Aaa by Moody; 21 refers to a C rating for both S&P and Moody. Ratings of 10 or below are considered investment-grade, and ratings above 10 are considered non-investment-grade. *Maturity* is the time to maturity of the bond in years. *Illiquidity* is the autocovariance of daily log bond price change in each month multiplied by −1 as defined in Bao et al. (2011). *Lag return* is the corporate bond return in the past month. *VaR (5%)* is the 5% value at risk of corporate bond return, defined as the second lowest monthly return over the past 36 months. Panel A reports the number of bond-month observations, mean, standard deviation, median, and percentiles of the variables. Panel B reports the time-series average of the cross-sectional correlation of the variables. All variables are winsorized each month at the 0.5% level. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

In Panel B of Table 1, we report the time-series average of the cross-sectional correlations of the variables. $\Delta CVOL$ and $\Delta PVOL$ have a moderate correlation of 0.58, indicating that the implied volatility changes of calls and puts might have a common component. The changes in implied volatilities are not correlated with implied volatility levels with a correlation of 0.08 and 0.11 for calls and puts, respectively. This suggests that innovations in implied volatility represent distinct information from the implied volatility level. This fact is important to note for later when we control for various volatility characteristics. Panel B also shows that the correlation between $\Delta CVOL - \Delta PVOL$ and $\Delta CVOL + \Delta PVOL$ is zero, suggesting that these two implied volatility variables carry different information. We observe that implied volatility is moderately correlated with *Rating* and *VaR* (5%) with a correlation of 0.50 and -0.47 , respectively. However, the correlations between changes in implied volatilities and various bond characteristics are much lower, ranging from -0.14 to 0.11 . It is less likely, therefore, that the predictability that we document later is related to the heterogeneity in bond characteristics.

3. Changes in Implied Volatility and the Cross-Section of Corporate Bond Returns

3.1. Bond Portfolios Sorted on Change in Call- or Put-Implied Volatility

For each month from July 2002 to August 2017, we form decile portfolios by sorting corporate bonds according to the change in the corresponding call- and put-implied volatilities, $\Delta CVOL$ and $\Delta PVOL$, of the underlying firms. Decile 1 contains bonds of firms with the largest decrease in implied volatilities in the previous month and decile 10 contains bonds of firms with the largest increase in implied volatility in the previous month. The portfolios are value-weighted using the prior month's bond market capitalization as weights. The holding period for the bonds is one month, and we rebalance the portfolios monthly.

We report the results for portfolios sorted on changes in implied volatilities of the call options, $\Delta CVOL$, in Panel A of Table 2. The returns are in percentage per month, and Newey–West *t*-statistics are reported in parenthesis below the returns. The average bond return in decile 1 is 0.95% and declines almost monotonically to 0.44% for bonds in decile 10. The difference in average raw returns between deciles 10 and 1 is -0.52% (*t*-statistic = -3.57). Thus, the negative relation between change in implied volatility and future bond return is both economically and statistically significant.

To examine whether the return spreads of the strategy can be explained by common risk factors in the

bond and stock markets, we calculate the alpha of the strategy spread using two factor models. We first use a bond factor model proposed by Bai et al. (2019), which consists of five bond market factors: excess bond market return (MKT_{bond}), downside risk factor (DRF), credit risk factor (CRF), liquidity risk factor (LRF), and reversal factor (REV).¹¹ We also consider stock market factors. We use a six-factor stock pricing model from Fama and French (2018). The six factors are excess stock market return (MKT), size factor (SMB), value factor (HML), profitability factor (RMW), investment factor (CMA), and momentum factor (MOM). Data on these stock factors are obtained from Ken French's website. Our second factor model is a joint bond + stock model that uses the five bond market factors and the six stock market factors.

We first examine the factor loadings of the decile portfolios on these factors. In unreported results, we find that, with a few exceptions, factor loadings in the bond + stock model of decile 1 are similar to those of decile 10. One big exception is that the loading on the MKT_{bond} factor is 0.66 (0.12) for decile 1 (10), leading to a negative (and statistically significant) loading on this factor for the 10–1 portfolio. The other two exceptions are loadings on the LRF and the REV factors. These are -0.37 (0.39) and -0.12 (0.31), respectively, for decile 1 (10) leading to positive (and statistically significant) loading on these factors for the 10–1 portfolio. Loadings of the 10–1 portfolio on MKT and MOM are also positive albeit smaller in magnitude. In general, therefore, the factor loadings of the 10–1 portfolio are positive. This also means that the alpha of the 10–1 portfolio is more negative than the raw return difference; and the bond + stock alpha is bigger (in absolute value) than the bond alpha. Thus, we find that the alphas of the 10–1 hedge portfolio from the bond factor model and from the bond + stock factor model are -0.58% (*t*-statistic = -2.53) and -0.90% (*t*-statistic = -4.94), respectively. The alphas also show a monotonic pattern across deciles. We conclude that the pattern in average returns across deciles is not explained by the common risk factor models.¹²

We also report several bond characteristics in each decile portfolio in the bottom part of Panel A of Table 2. The bond characteristics that we include are bond market beta (*Beta*), *Size*, *Maturity*, *Rating*, and *Illiquidity*. *Beta* is the regression coefficient of returns on bond market factor in the bond factor model. As noted earlier, *Beta* of decile 1 is higher than that of decile 10; however, the betas do not show a clear pattern across the other deciles. Similarly, we find no evidence that other bond characteristics are driving the cross-section pattern we document. *Size* is similar across deciles. Bonds in extreme deciles 1 and 10 have shorter maturity, lower rating, and are less liquid

Table 2. Portfolio of Bonds Sorted by $\Delta CVOL$ or $\Delta PVOL$

	1	2	3	4	5	6	7	8	9	10	10–1
Panel A: Portfolios sorted by $\Delta CVOL$											
Average return	0.95 (4.29)	0.72 (4.70)	0.63 (5.11)	0.51 (4.81)	0.49 (4.53)	0.49 (4.28)	0.47 (3.83)	0.45 (3.89)	0.48 (3.35)	0.44 (2.28)	–0.52*** (–3.57)
Bond alpha	0.23 (1.75)	0.21 (1.61)	0.02 (0.16)	0.01 (0.06)	–0.08 (–0.70)	–0.11 (–0.80)	–0.22 (–1.26)	–0.16 (–1.01)	–0.35 (–1.91)	–0.35 (–2.28)	–0.58*** (–2.53)
Bond + stock alpha	0.45 (4.40)	0.41 (4.14)	0.25 (5.80)	0.21 (4.10)	0.12 (2.66)	0.13 (2.54)	0.03 (0.59)	0.07 (1.07)	–0.14 (–1.92)	–0.45 (–3.20)	–0.90*** (–4.94)
Beta	0.59	0.55	0.56	0.51	0.47	0.65	0.68	0.63	0.76	0.49	
Size	12.47	12.41	12.48	12.43	12.46	12.43	12.47	12.52	12.47	12.42	
Maturity	7.80	8.44	8.73	8.73	8.71	8.78	8.72	8.65	8.58	7.97	
Rating	10.76	8.76	7.97	7.67	7.58	7.53	7.56	7.85	8.46	10.31	
Illiquidity	1.27	1.10	1.04	1.02	0.99	0.98	1.01	1.05	1.09	1.42	
Panel B: Portfolios sorted by $\Delta PVOL$											
Average return	0.96 (4.39)	0.63 (4.85)	0.65 (5.14)	0.58 (5.17)	0.54 (4.52)	0.44 (4.25)	0.44 (3.62)	0.48 (4.14)	0.45 (2.75)	0.46 (2.35)	–0.50*** (–3.51)
Bond alpha	0.24 (1.89)	0.07 (0.59)	0.13 (1.00)	0.06 (0.52)	–0.05 (–0.40)	–0.10 (–0.88)	–0.23 (–1.15)	–0.18 (–1.30)	–0.34 (–1.49)	–0.40 (–2.50)	–0.64*** (–2.72)
Bond + stock alpha	0.39 (3.92)	0.30 (6.89)	0.34 (4.21)	0.25 (4.85)	0.20 (3.89)	0.10 (2.20)	0.04 (0.73)	0.07 (1.32)	–0.07 (–0.81)	–0.50 (–3.23)	–0.89*** (–4.40)
Beta	0.48	0.59	0.54	0.46	0.54	0.57	0.72	0.69	0.80	0.58	
Size	12.47	12.44	12.41	12.44	12.49	12.51	12.45	12.48	12.50	12.42	
Maturity	7.72	8.46	8.61	8.70	8.79	8.82	8.70	8.68	8.60	7.98	
Rating	10.80	8.77	8.00	7.64	7.54	7.50	7.58	7.87	8.44	10.38	
Illiquidity	1.30	1.08	1.01	1.02	0.99	0.99	1.02	1.04	1.08	1.42	

Notes. At the end of each month, we sort bonds into deciles. Decile 1 is the portfolio with the lowest changes in implied volatilities, and decile 10 is the portfolio with the highest changes in implied volatilities. The implied volatility is calculated from at-the-money options with 365 days to maturity, and the change is calculated as the difference over the last month. We use changes in implied volatilities of call options ($\Delta CVOL$) in Panel A and that of put options ($\Delta PVOL$) in Panel B. The portfolios are held for one month and rebalanced monthly. Portfolios are value-weighted using the prior month's bond market capitalization as weights. We report the average returns of the deciles as well as portfolio alphas. Alphas are calculated from a bond model and a bond + stock model. The bond model uses five factors (bond market, downside risk, credit risk, liquidity risk, and reversal) from Bai et al. (2019). The stock market factors are the six factors (market, size, value, investment, profitability, and momentum) from Fama and French (2018). The bond + stock model combines the five bond factors and the six stock market factors. All returns and alphas are in percentage per month. Newey–West adjusted t -statistics are reported in parenthesis below returns/alphas. The last column reports the returns and alphas for the 10–1 portfolio. Finally, we also report a few bond characteristics for each decile. These characteristics are bond market factor beta, size, rating, maturity, and illiquidity. Details on construction of these variables are provided in Table 1. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

compared with bonds in other deciles. However, the differences in these characteristics between deciles 1 and 10 are not significant. This suggests that the relation between the changes in implied volatilities and future corporate bond return is unlikely to be explained by these variables. Nevertheless, because there is some relation between these characteristics and portfolio sorts, we explore the impact of bond characteristics on portfolio returns in greater detail in the next section.

We report the results for portfolios sorted on changes in implied volatilities of the put options, $\Delta PVOL$ in Panel B of Table 2. Sorting by $\Delta PVOL$ yields similar results as those from sorting by $\Delta CVOL$. The raw average return of the decile portfolios decreases from 0.96% in decile 1 to 0.46% in decile 10, leading to a return spread of –0.50% (t -statistic = –3.51). The alphas of the 10–1 portfolio from the bond factor model and the bond + stock factor model are –0.64% (t -statistic = –2.72) and –0.89% (t -statistic =

–4.40), respectively. Similar to the results for $\Delta CVOL$ -sorted bonds, characteristics do not exhibit clear patterns from decile 1 to decile 10 for portfolio sorts on $\Delta PVOL$ either.

3.2. Bond Portfolios Sorted on the Common Component of Call- and Put-Implied Volatility Changes

It is interesting to note the consistency of our results for sorts involving $\Delta CVOL$ and $\Delta PVOL$. This is in contrast to An et al. (2014), who find that $\Delta CVOL$ ($\Delta PVOL$) predicts future stock returns positively (negatively). An et al. (2014) posit that change in implied volatility carries information about fundamental news of the underlying firm. Our results tend to suggest that the change in implied volatility could also convey information about default risk of the underlying firm. The volatility news is likely to be captured by the common component of $\Delta CVOL$ and $\Delta PVOL$.

In other words, an increase in $\Delta CVOL$ could be either due to good fundamental news or about increasing risk, whereas an increase in $\Delta PVOL$ could be either due to bad fundamental news or about increasing risk. One simple way to extract the common component of $\Delta CVOL$ and $\Delta PVOL$, that is, related to risk only, is to take the sum $\Delta CVOL + \Delta PVOL$. In this case, the difference $\Delta CVOL - \Delta PVOL$ is likely to be related to fundamental news. Recall that Panel B of Table 1 shows zero correlation between $\Delta CVOL - \Delta PVOL$ and $\Delta CVOL + \Delta PVOL$, indicating that these two implied volatility variables carry different information.

Using Merton (1973), one can view the fundamental news proxy, $\Delta CVOL - \Delta PVOL$, as affecting the asset value and thereby affecting the value of claims (stocks and bonds) on it—the delta effect. $\Delta CVOL - \Delta PVOL$ is the main variable used by An et al. (2014), consistent with their claim that news about fundamentals, extracted from options, predicts stock price changes. The news about changing risk, $\Delta CVOL + \Delta PVOL$, in this framework is then the vega effect on stocks and bonds. Whereas the main determinant for (change in) bond prices in the Merton (1973) model is asset volatility, Choi and Richardson (2016) show that implied volatility and firms' asset volatility are closely linked. Thus, contingent claim pricing intuition would suggest that increase in implied volatility would reduce bond prices (as bonds are short put positions on the asset value).

We repeat the portfolios sorts in Table 3 with $\Delta CVOL + \Delta PVOL$ (Panel A) and $\Delta CVOL - \Delta PVOL$ (Panel B) as the sorting variables. Panel A shows that $\Delta CVOL + \Delta PVOL$ sorts produce a spread in returns and alphas that is even bigger than those reported in

Table 2; the 10–1 return is -0.60% , and the bond + stock alpha is -0.98% per month. In contrast, sorts on $\Delta CVOL - \Delta PVOL$ produce a spread in bond returns that is economically and statistically insignificant.¹³

Do the bond investors underreact or react with a delay to the information provided by $\Delta CVOL + \Delta PVOL$? To answer this, we look at the same sorts but analyze formation-period returns. We find that the formation-month 10–1 return is -2.13% (t -statistic = -6.40) and the bond + stock alpha is -1.87% (t -statistic = -6.96) per month for $\Delta CVOL + \Delta PVOL$ sorts (sorts on $\Delta CVOL - \Delta PVOL$ continue to produce insignificant spreads in the formation-period month). Thus, the evidence is more suggestive of underreaction rather than a completely delayed reaction.

Because good fundamental news is good for both stocks and bonds, one might expect that even the $\Delta CVOL - \Delta PVOL$ sorts should produce a positive spread in bond returns. Note, however, that bonds have a limited upside in contrast to stocks (Hong and Sraer 2013). The upside is even more limited for high-rated bonds. The conjecture is, however, more plausible for low-rated bonds. In unreported results, we do find that $\Delta CVOL - \Delta PVOL$ sorts produce a spread of 0.36% (t -statistic = 1.16) in the postformation month and a spread of 1.05% (t -statistic = 2.30) in the formation month for the subsample of non-investment-grade bonds.

Yet another possibility is that fundamental news information in $\Delta CVOL - \Delta PVOL$ is masked by the increase in risk information in $\Delta CVOL + \Delta PVOL$. To check this, we do a 5×5 double sort on $\Delta CVOL - \Delta PVOL$ and $\Delta CVOL + \Delta PVOL$. In unreported results, we find that spread in returns from $\Delta CVOL - \Delta PVOL$ sorts is 0.30% (t -statistic = 2.22) when $\Delta CVOL +$

Table 3. Return and Alphas of Portfolio of Bonds Sorted by $\Delta CVOL + \Delta PVOL$ or $\Delta CVOL - \Delta PVOL$

	1	2	3	4	5	6	7	8	9	10	10–1
Panel A: Portfolios sorted by ($\Delta CVOL + \Delta PVOL$)											
Average return	0.98 (4.50)	0.64 (4.61)	0.61 (4.83)	0.65 (5.16)	0.48 (4.40)	0.51 (3.95)	0.51 (4.74)	0.39 (3.05)	0.45 (3.39)	0.38 (1.85)	-0.60^{***} (-3.62)
Bond alpha	0.24 (1.93)	0.17 (1.28)	0.01 (0.09)	0.07 (0.53)	-0.10 (-0.71)	-0.10 (-0.71)	-0.13 (-1.08)	-0.34 (-1.55)	-0.22 (-1.41)	-0.47 (-3.09)	-0.71^{***} (-3.50)
Bond + stock alpha	0.42 (4.07)	0.37 (4.68)	0.25 (4.46)	0.28 (5.07)	0.13 (2.76)	0.16 (2.24)	0.09 (1.77)	-0.07 (-1.06)	-0.03 (-0.34)	-0.57 (-3.74)	-0.98^{***} (-5.57)
Panel B: Portfolios sorted by ($\Delta CVOL - \Delta PVOL$)											
Average return	0.67 (3.07)	0.52 (4.22)	0.50 (4.22)	0.50 (4.06)	0.55 (4.60)	0.47 (4.16)	0.50 (4.08)	0.49 (4.34)	0.57 (4.33)	0.80 (4.00)	0.13 (1.03)
Bond alpha	0.04 (0.39)	-0.11 (-0.96)	-0.11 (-0.84)	-0.16 (-1.08)	-0.11 (-0.79)	-0.15 (-1.06)	-0.12 (-0.77)	-0.14 (-0.94)	-0.06 (-0.45)	0.10 (0.86)	0.06 (0.42)
Bond + stock alpha	0.21 (2.36)	0.08 (1.06)	0.09 (1.81)	0.08 (1.42)	0.13 (2.62)	0.07 (1.51)	0.12 (2.02)	0.08 (1.64)	0.13 (2.03)	0.15 (1.37)	-0.05 (-0.37)

Notes. This table presents portfolio sort results for bonds sorted by the sum ($\Delta CVOL + \Delta PVOL$) (Panel A) and the difference ($\Delta CVOL - \Delta PVOL$) (Panel B) of call and put implied volatility changes. Portfolios are sorted as in Table 2. This table shows the returns and alphas on the decile portfolios and the 10–1 portfolio. All returns and alphas are in percentage per month. Newey–West adjusted t -statistics are reported in parenthesis below returns/alphas. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

$\Delta PVOL$ is the lowest. Thus, $\Delta CVOL - \Delta PVOL$ predicts higher future bond returns only when there is no countervailing force of increase in default risk.

Some readers suggest that an alternative way to display our results would be via a double sort on $\Delta CVOL$ and $\Delta PVOL$. In unreported results, we do an independent 3×3 sort on these two changes in implied volatility. We find that the (1, 1) portfolio, in which both implied volatilities decrease the most, has a bond + stock alpha of 0.36% per month (comparable to a bond + stock alpha of 0.42% in decile 1 of Panel A of Table 3). Similarly, the (3, 3) portfolio, in which both implied volatilities increase the most, has a bond + stock alpha of -0.56% per month (comparable to a bond + stock alpha of -0.57% in decile 10 of Panel A of Table 3). We opt for simple decile sorts in the rest of the paper for expositional purposes.

Henceforth, we relabel $(\Delta CVOL + \Delta PVOL)/2$ as $\Delta ImpVOL$ and consider this as our main forecasting variable.¹⁴ We also consider $\Delta ImpVOL$ as indicative of future changes in volatility (see Christensen and Prabhala 1998 for an early study on the use of implied volatility to predict future volatility) and as indicative of future changes in default risk (Shumway 2001, Campbell and Taksler 2003). We provide further evidence on these issues in Section 4.

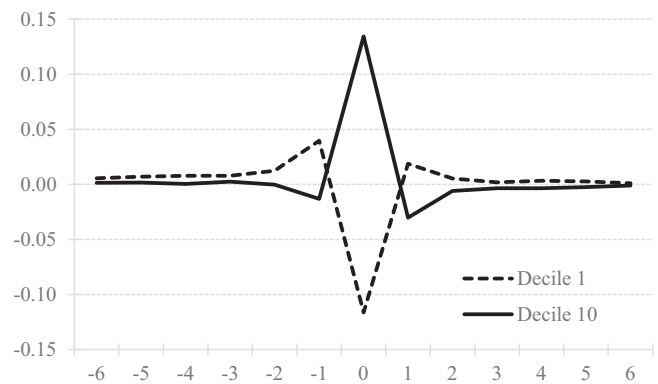
Figure 1 plots $\Delta ImpVOL$ for an interval of six months around the portfolio formation month for deciles 1 and 10. Apart from the changes in the formation month, by construction, we do not observe large changes in implied volatility in the other months. This means that implied volatility stays at relatively high (low) levels for decile 10 (1) after the shock in the formation month. Online Appendix Table A1 shows that the predictability declines rapidly.

3.3. Control for Bond Characteristics: Double Portfolio Sorts

The univariate portfolio sort in Section 3.1 shows a strong negative relation between $\Delta ImpVOL$ and future bond returns. However, it is possible that $\Delta ImpVOL$ is correlated with bond characteristics, and thus, we are picking up the relation between bond returns and these characteristics. Indeed, prior studies show that bond characteristics, such as maturity, coupon, age, and ratings, can explain the cross-section of corporate bond returns (see, for example, Gebhardt et al. 2005a, Bai et al. 2019). Portfolio characteristics reported in Table 2 also show that there is some relation between sorts and some characteristics. Therefore, to assess the robustness of our results, we control for characteristics in portfolio sorts.

We construct conditional double-sorted portfolios. We first sort bonds into quintile portfolios based on a single characteristic. Following Bai et al. (2019), we

Figure 1. $\Delta ImpVOL$ Around Portfolio Formation Month



Notes. This figure shows change of implied volatility, $\Delta ImpVOL$, in the 1st and 10th deciles sorted by $\Delta ImpVOL$. The portfolios are formed at month t , and the graph shows $\Delta ImpVOL$ from month $t - 6$ to month $t + 6$. The sample period is from July 2002 to August 2017.

choose *Size*, *Maturity*, *Rating*, *Illiquidity*, and *Lag return* as control characteristics. In sorting by *Rating*, we do not sort into equal-sized bins, but opt for sorting based on more intuitive classifications. In particular, the five quintiles contain bonds rated AAA to AA–, A+ to A–, BBB+ to BBB–, BB+ to BB–, and below, respectively. Thus, the first three quintiles contain investment-grade bonds, and quintiles four and five contain non-investment-grade bonds.

Within each characteristic quintile, we further sort bonds into five quintiles based on $\Delta ImpVOL$. We calculate the bond + stock alpha of the 5–1 hedge portfolio that is long in bonds with the highest $\Delta ImpVOL$ and short in bonds with the lowest $\Delta ImpVOL$. This long–short portfolio alpha is calculated for each of the characteristic quintiles and is similar to the approach in Chung et al. (2019). This approach not only shows that the returns/alphas are robust after controlling for characteristic, but also shows the variation, if any, in the magnitude of profitability across quintiles of characteristics.

Results in Panel A of Table 4 show that, although there is some variation across the size quintile portfolios, 5–1 portfolio alphas are significant for quintiles 2 to 5. Whereas the magnitude of the spread for the smallest size quintile is similar to that of the other four quintiles, the spread for the smallest size quintile is not statistically significant. These results indicate that the effect of $\Delta ImpVOL$ is not concentrated among smaller or bigger bonds. This result is not surprising as Table 2 shows little variation in size across decile portfolios.

For portfolios sorted by maturity, we find that 5–1 portfolio alphas exhibit an increasing pattern (in absolute terms) from quintile 1 of short-maturity bonds to quintile 5 of long-maturity bonds. For example, the

Table 4. Alphas of 5–1 Portfolio of Bonds Sorted by $\Delta ImpVOL$ Controlling for Bond and Volatility Characteristics

	1	2	3	4	5	
Panel A: 5–1 alphas for each bond characteristic quintile						
Size	−0.58 (−1.39)	−0.69*** (−5.18)	−0.67*** (−5.25)	−0.61*** (−4.86)	−0.61*** (−5.00)	
Maturity	0.09 (0.62)	−0.48*** (−4.55)	−0.92*** (−5.81)	−0.77*** (−4.98)	−1.32*** (−4.12)	
Rating	−0.09 (−1.44)	−0.73*** (−3.20)	−0.63*** (−3.63)	−0.51** (−2.10)	−0.69** (−2.14)	
Illiquidity	−0.75*** (−5.16)	−0.28*** (−3.77)	−0.52*** (−3.95)	−0.79*** (−4.47)	−1.19*** (−3.31)	
Lag return	−1.03*** (−2.88)	−0.37*** (−3.40)	−0.27*** (−3.71)	−0.32*** (−4.12)	−0.56** (−2.50)	
Panel B: 5–1 alphas for each volatility quintile						
Bond Vol	−0.25*** (−3.25)	−0.47*** (−3.91)	−0.73*** (−3.58)	−1.05*** (−5.11)	−1.22*** (−3.21)	
Bond IdioVol	−0.17*** (−3.29)	−0.36*** (−3.58)	−0.55*** (−4.07)	−0.98*** (−4.16)	−1.22*** (−3.12)	
ImpVOL	−0.10 (−1.45)	−0.15*** (−2.76)	−0.15 (−1.61)	−0.22** (−2.41)	−0.78*** (−3.12)	
Stock IdioVol	−0.13** (−2.04)	−0.17* (−1.67)	−0.28** (−2.59)	−0.34** (−2.18)	−0.76*** (−2.86)	
VIX Beta	−0.93*** (−4.00)	−0.55*** (−3.92)	−0.42*** (−4.43)	−0.51*** (−5.31)	−1.21*** (−4.31)	
	1	2	3	4	5	5–1
Panel C: Alphas of each $\Delta ImpVOL$ quintile controlling for bond characteristics						
Size	0.31 (3.90)	0.27 (5.04)	0.16 (4.93)	0.02 (0.45)	−0.32 (−3.71)	−0.63*** (−4.37)
Maturity	0.39 (6.20)	0.25 (5.04)	0.15 (3.20)	0.01 (0.30)	−0.29 (−3.03)	−0.68*** (−4.77)
Rating	0.32 (5.22)	0.27 (5.45)	0.16 (3.08)	0.10 (1.90)	−0.28 (−3.03)	−0.60*** (−4.92)
Illiquidity	0.41 (6.18)	0.29 (5.41)	0.15 (3.08)	0.02 (0.36)	−0.30 (−2.89)	−0.71*** (−4.82)
Lag return	0.33 (6.75)	0.32 (4.37)	0.13 (2.82)	−0.01 (−0.30)	−0.18 (−2.16)	−0.51*** (−4.57)
Panel D: Alphas of each $\Delta ImpVOL$ controlling for volatility characteristics						
Bond Vol	0.43 (6.51)	0.37 (4.67)	0.15 (2.78)	−0.03 (−0.54)	−0.32 (−2.74)	−0.74*** (−4.88)
Bond IdioVol	0.36 (5.95)	0.32 (5.54)	0.13 (2.55)	−0.04 (−0.89)	−0.30 (−2.73)	−0.65*** (−4.84)
Stock ImpVOL	0.27 (4.61)	0.18 (3.75)	0.08 (1.49)	0.07 (1.80)	−0.01 (−0.19)	−0.28*** (−4.50)
Stock IdioVol	0.26 (6.39)	0.28 (5.11)	0.11 (2.15)	0.00 (0.10)	−0.07 (−1.14)	−0.34*** (−5.01)
VIX Beta	0.41 (6.71)	0.28 (4.89)	0.16 (2.81)	0.01 (0.27)	−0.31 (−3.06)	−0.72*** (−5.27)

Notes. At the end of each month, we conditionally double sort the bonds into 5×5 quintiles. In Panels A and C, we first sort bonds into quintile portfolios based on a characteristic, which is size, maturity, rating, illiquidity, or bond return in the past month. Details on the construction of these variables are in Table 1. In Panels B and D, we first sort bonds into quintile portfolios based on a volatility characteristic, which is bond volatility, bond idiosyncratic volatility, stock implied volatility, stock idiosyncratic volatility, and VIX beta. Bond volatility (*BondVol*) is calculated as the standard deviation of daily bond returns within each month. Bond idiosyncratic volatility (*Bond IdioVol*) is the standard deviation of bond return residuals estimated from the time-series regression with five Fama and French (2015) factors and change in VIX as volatility risk factor. Stock implied volatility (*ImpVOL*) is the average of the call and put at-the-money implied volatility with 365 days of expiration. Stock idiosyncratic volatility (*Stock IdioVol*) is the standard deviation of stock return residuals estimated from the time-series regression with five stock factors and the volatility risk factor. *VIX beta* is the regression coefficient on change in VIX estimated from the same time-series regression used to estimate bond idiosyncratic volatility. Within each characteristic quintile, we further sort bonds into five quintiles based on $\Delta ImpVOL$ ($\equiv (\Delta CVOL + \Delta PVOL)/2$). We calculate the bond + stock model alpha of the 5 – 1 hedge portfolio that is long in bonds with the highest changes in implied volatilities and short in bonds with the lowest changes in implied volatilities. The factor model is the same as that in Table 2. This alpha is calculated for hedge portfolios in each of the characteristic quintiles. Panels A and B report these alphas together with their Newey–West adjusted *t*-statistics in parenthesis. In Panels C and D, we average the return of each $\Delta ImpVOL$ quintile across the five characteristic portfolios and then report alphas of these quintile portfolios and that of the 5 – 1 portfolio. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

5–1 alpha for $\Delta ImpVOL$ sorted portfolios is 0.09% and –1.32% for quintiles 1 and 5, respectively. Similar to our findings, Bai et al. (2019) find that the bond return spread between the top and bottom downside risk quintiles is larger for long-term bonds.

Similar to prior studies, such as Jostova et al. (2013) and Bai et al. (2019), we find stronger return predictability for low-rated bonds (quintile 5 of rating) than for high-rated bonds (quintile 1 of rating). Note that we include credit risk factor in our factor model. Thus, the alpha differences across rating categories in Panel A are compensation beyond what has been accounted for by risk models. We find that the 5–1 alpha for $\Delta ImpVOL$ sorted portfolios is –0.69% for low-rated bonds, which is around seven times the magnitude of –0.09% for high-rated bonds. In terms of the Merton (1973) model, high-rated bonds are out-of-the-money put options and, therefore, essentially risk-free. Therefore, $\Delta ImpVOL$ carries little information for these bonds, which is what we find.

Bai et al. (2019), Bongaerts et al. (2017), and Lin et al. (2011) find that liquidity (risk) is priced in corporate bonds. Whereas we see bond predictability in all quintiles of liquidity, the 5–1 alpha for $\Delta ImpVOL$ sorted portfolios is higher (in absolute magnitude) at –1.19% for more illiquid bonds than it is for more liquid bonds at –0.75%.

We also find that $\Delta ImpVOL$ predictability is higher for past bond losers (5–1 alpha of –1.03%) than for past bond winners (5–1 alpha of –0.56%). Because prior literature finds evidence of corporate bond momentum (Gebhardt et al. 2005b, Jostova et al. 2013), it is possible that our signal of $\Delta ImpVOL$ counteracts the effect of momentum for some bonds. In fact, in unreported results, we find a positive alpha of 0.13% (albeit statistically insignificant) for the portfolio of past bond winners, and high $\Delta ImpVOL$ indicates that the signal of past bond returns outweighs the signal of increase in implied volatility for these bonds.

Overall, we see that the predictability is related to some, but not all, characteristics. Whereas we find evidence of predictability across most of our double-sorted portfolios, it is still possible that we have picked up the differences in bond characteristics. In Section 3.5, we control for those characteristics using regression analysis.

Our approach of reporting 5–1 alphas for $\Delta ImpVOL$ sorts for each characteristic quintile is a strong test of predictability across characteristic quintiles. A less strong but, nevertheless, intuitive alternative approach is to calculate 5–1 alphas for $\Delta ImpVOL$ after controlling for each characteristic. This alternative approach is frequently used in the literature (see, for example, Ang

et al. 2006, Bai et al. 2019, Chung et al. 2019). In particular, we perform the same conditional sorts—first sort on a characteristic and then within each characteristic quintile and further sort into quintiles based on $\Delta ImpVOL$. We average the return for each $\Delta ImpVOL$ quintile across the five characteristic portfolios. This approach produces portfolios that vary in $\Delta ImpVOL$ but have similar bond characteristics.

We report the returns/alphas for each $\Delta ImpVOL$ quintile as well as the alpha for the 5–1 hedge portfolio in Panel C of Table 4. We find that the 5–1 alphas for $\Delta ImpVOL$ sorted portfolios show little variation across characteristics. This result is consistent with the descriptive statistics of the portfolios in Table 2, which show little variation across characteristics. All the alphas are also statistically significant. The magnitude of alphas in Panel C is roughly half of that in Table 3 as Panel C uses quintile sorts (controlling for characteristics), whereas Table 3 uses decile sorts (unconditional on characteristics).

3.4. Control for Volatility Characteristics: Double Portfolio Sorts

Chung et al. (2019) find that bonds with high volatility betas or low idiosyncratic bond volatility have higher expected returns. Our main volatility sorting variable is different from that used by these authors. Nevertheless, there could be correlations among different volatility variables, and hence, our results could potentially be driven by the exposure to volatility risk and/or bond volatility. Therefore, in this section, we test whether volatility-related characteristics can explain our results.

We use five different volatility-related variables as control variables: bond volatility, bond idiosyncratic volatility, stock implied volatility, stock idiosyncratic volatility, and VIX beta. In particular, *Bond Vol* is calculated as the standard deviation of daily bond returns within each month.¹⁵ We calculate the *Bond IdioVol* as the standard deviation of bond return residuals, estimated from the time-series regression with five Fama and French (2015) factors and change in VIX as volatility risk factor. *ImpVOL* is the stock-implied volatility calculated as before as the average of the call and put at-the-money implied volatility with 365 days of expiration. Similar to bond idiosyncratic volatility, *Stock IdioVol* is the standard deviation of stock return residuals estimated from the time-series regression with the same factor model as that for bonds. Finally, *VIX Beta* is the regression coefficient of the change in VIX estimated from the same time series regression as that used to calculate bond idiosyncratic volatility.

We follow the same conditional sorting procedure as that in Section 3.3. In particular, we first sort bonds into quintile portfolios based on a volatility characteristic. Within each volatility characteristic quintile, we further sort bonds into five quintiles based on $\Delta ImpVOL$. We calculate the stock + bond model alpha of the 5–1 hedge portfolio that is long in bonds with the highest change in $\Delta ImpVOL$ and short in bonds with the lowest change in $\Delta ImpVOL$.

Panel B of Table 4 shows that the effect of changes in implied volatilities on bond return predictability is not concentrated in any particular quintile of bonds with certain volatility characteristics. Almost all 5–1 alphas are statistically significant (with only three exceptions across the two panels); most are significant at the 1% level.

At the same time, the effect of $\Delta ImpVOL$ is related to volatility-related characteristics. We find that 5–1 alphas are more negative for bonds with higher bond volatility, higher idiosyncratic bond volatility, higher stock implied volatility, and higher stock idiosyncratic volatility. The predictability of changes in implied volatilities does not show a clear pattern in the quintile portfolios sorted by *VIX Beta*.

We also calculate returns of $\Delta ImpVOL$ sorted portfolios controlling for the volatility characteristics in a manner similar to that explained toward the end of Section 3.3. Panel D of Table 4 shows that 5–1 alphas for $\Delta ImpVOL$ sorted portfolios show little variation across volatility characteristics. Overall, our results show that the effect of changes in implied volatilities is related to stock and bond volatility but cannot be fully explained by them.

3.5. Fama-MacBeth Regressions

We next examine the cross-sectional relation between changes in implied volatilities and bond returns at the individual bond level using Fama and MacBeth (1973) regressions. We estimate the regression across all bonds in each month and then report the cross-sectional average of the coefficients. We calculate Newey–West *t*-statistics with six lags and report them below the coefficients. All independent variables in all regressions are winsorized each month at the 0.5% level.

We report results in Table 5 for univariate regressions on $\Delta ImpVOL$ as well as regressions with controls. The average coefficient on $\Delta ImpVOL$ is -0.051 with a *t*-statistic of -4.55 . To gauge the economic magnitude of these coefficients, note first that the difference in $\Delta ImpVOL$ is 11% as one goes from the 1st to the 10th decile of bonds. Therefore, based on this coefficient estimate, bonds in the 1st decile of *ImpVOL* outperform bonds in the 10th decile of *ImpVOL* by $0.051 \times 11\% = 0.56\%$, *ceteris paribus*.

Model (2) in Table 5 controls for additional variables. The control variables are the same ones that we

use in Table 4, namely, *Size*, *Rating*, *Maturity*, *Illiquidity*, and *Lag return*. In addition, we also include *ImpVOL* and *VaR* (5%) as control variables. Table 4 shows that profitability of our strategy is related to rating, maturity, and illiquidity of bonds. Bai et al. (2019) find that corporate bonds with higher downside risk, measured by *VaR* (5%), earn significantly higher returns than bonds with lower downside risk. In addition, Bali et al. (2021) also find that the previous month's bond return has strong ability to predict future bond returns in the cross-section. We include lagged stock returns based on evidence in Chordia et al. (2017) and Gebhardt et al. (2005b). We also include implied volatility as a control to make sure that the negative relation between changes in implied volatilities and future bond returns is not driven by the level of implied volatility.

Consistent with the findings in Bai et al. (2019), we find that the coefficients on *Lag bond return* and *VaR* (5%) are both strongly statistically significant. Consistent with the findings in Chordia et al. (2017), we find that the coefficient on *Lag stock return* is positive and strongly statistically significant. The coefficients on *ImpVOL* are negative in all regressions, but none of them is statistically significant. Together with the low correlation between changes in implied volatilities and implied volatility level in Table 1, the results indicate that the effect of implied volatility level is very different from that of changes in implied volatilities. An et al. (2014) report similar differences between the impact of levels and changes in implied volatilities on stock returns. Coefficients on *Size* and *Illiquidity* are statistically significant in some specifications but not all, whereas coefficients on *Rating* and *Maturity* are not statistically significant in any specification that we explore. The average coefficient on $\Delta ImpVOL$ is -0.043 (*t*-statistic = -5.28) and barely changes from its univariate counterpart.

In Model (3) of Table 5, we test whether the effect of $\Delta ImpVOL$ is robust after controlling for the volatility-related variables used in Table 4. In particular, we include bond idiosyncratic volatility, stock idiosyncratic volatility, and *VIX beta*. We follow Chung et al. (2019) in the construction of these variables. We find that the coefficient on $\Delta ImpVOL$ is statistically significant in Model (3) also with coefficient estimate of -0.039 (*t*-statistic = -4.55). Consistent with Chung et al. (2019), we find the coefficient of bond idiosyncratic volatility to be positive and significant. However, after including changes in implied volatilities in the regression, we find that coefficients on *VIX beta* and idiosyncratic stock volatility become insignificant.

Bai et al. (2019) provide evidence of a positive cross-sectional relation between the level of the physical measure of bond volatility and future bond returns, but to the best of our knowledge, no study explores

Table 5. Fama–MacBeth Regressions for the Cross-Section of Bond Returns

	(1)	(2)	(3)	(4)	(5)
<i>Intercept</i>	0.005*** (4.56)	0.006*** (2.91)	−0.005 (−1.65)	−0.006* (−1.81)	−0.005 (−1.51)
$\Delta ImpVOL$	−0.051*** (−4.55)	−0.043*** (−5.28)	−0.039*** (−4.55)	−0.040*** (−4.63)	−0.047*** (−4.89)
<i>ImpVOL</i>		−0.001 (−0.17)	−0.004 (−0.94)	−0.003 (−0.67)	−0.002 (−0.35)
<i>Size</i>		−0.036*** (−2.94)	0.052** (2.59)	0.057*** (2.85)	0.055** (2.60)
<i>Rating</i>		0.012 (0.80)	0.009 (0.70)	0.012 (0.98)	0.012 (0.89)
<i>Maturity</i>		0.008 (1.21)	0.000 (0.73)	0.000 (0.74)	0.000 (1.01)
<i>Illiquidity</i>		−0.001*** (−3.14)	0.019 (0.90)	0.018 (0.87)	0.027 (1.27)
<i>Lag bond return</i>		−0.134*** (−8.12)	−0.092*** (−4.75)	−0.092*** (−4.82)	−0.112*** (−5.94)
<i>VaR (5%)</i>		−0.068*** (−3.11)	−0.052*** (−2.76)	−0.050*** (−2.73)	−0.044*** (−2.69)
<i>Lag stock return</i>		0.070*** (15.58)	0.080*** (14.83)	0.079*** (15.40)	0.076*** (13.04)
<i>Bond IdioVol</i>			0.117** (2.20)	0.122** (2.02)	0.081 (1.35)
<i>Stock IdioVol</i>			0.039 (1.17)	0.007 (0.20)	−0.016 (−0.38)
<i>VIX Beta</i>			0.014 (0.38)	0.010 (0.32)	0.019 (0.55)
$\Delta Bond Vol$				−0.044 (−1.39)	0.014 (0.51)
$\Delta Stock Vol$				0.040 (1.43)	0.059* (1.80)
<i>Bond Abn. Volume</i>					0.034** (2.10)
<i>Stock Abn. Volume</i>					−0.016 (−0.89)
Adjusted R^2	0.016	0.236	0.297	0.306	0.297
Observations	886,613	551,756	288,898	287,874	211,656

Notes. This table presents time-series averages of the monthly Fama and MacBeth (1973) regression coefficients and their corresponding Newey–West adjusted t -statistics. Model (2) controls for bond and stock characteristics. Model (3) adds volatility-related variables, Model (4) adds change in bond realized volatility and change in stock realized volatility, and Model (5) adds abnormal bond volume and abnormal stock volume as additional control variables. All independent variables are winsorized each month at the 0.5% level. “Adjusted R^2 ” is the average adjusted R^2 across months, and “Observations” is the total number of observations. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

cross-sectional relation between the change in physical measures of volatility. Accordingly, in Model (4) of Table 5, we add two variables related to change in realized volatility of bond and stock returns. We find that neither of these variables predicts future bond returns. The coefficient on $\Delta ImpVOL$ remains relatively unchanged at -0.040 (t -statistic = -4.63).¹⁶

Finally, in another attempt to control for illiquidity, we use abnormal trading volume. Gervais et al. (2001) provide evidence of a significantly positive cross-sectional relation between abnormal trading volume and future stock returns, that is, low abnormal trading volume leads to low future stock returns. To ensure that the negative relation between $\Delta ImpVOL$ and future bond returns is not driven by the low trading volume of stocks and/or bonds of the underlying

firm, we control for abnormal trading volume of both bonds and stocks (calculated using the procedure of Gervais et al. 2001) in Model (5) of Table 5. We find that bond abnormal volume predicts future bond returns, whereas stock abnormal volume does not. More importantly for us, the coefficient on $\Delta ImpVOL$ remains negative at -0.047 and statistically significant (t -statistic = -4.89). Overall, the evidence in Table 5 suggests that change in implied volatility subsumes a large portion of information in other bond and stock variables.

We refer the interested reader also to Online Appendix A, which contains a host of other robustness checks. These relate to using options of different moneyness, maturity, different bonds, additional controls, and subperiod evidence.

4. The Information Content of $\Delta ImpVOL$

So far, we establish a robust relation between $\Delta ImpVOL$ and future bond returns. Broadly speaking, these results show that bond prices at the time of portfolio formation do not fully incorporate information contained in options. In this section, we investigate the nature of information in $\Delta ImpVOL$ that leads to future negative bond returns. We first show that $\Delta ImpVOL$ predicts future changes in default risk. We then show that $\Delta ImpVOL$ also predicts CDS spread change.

4.1. Predicting Future Change in Default Risk from $\Delta ImpVOL$

Equity volatility is shown to have power in bankruptcy prediction models (see, for example, Shumway 2001, Chava and Jarrow 2004, Campbell et al. 2008). Similarly, Campbell and Taksler (2003) and Zhang et al. (2009) show that equity volatility is useful for explaining cross-sectional variation in bond yields. Given that $\Delta ImpVOL$ signals an increase in future volatility (Christensen and Prabhala 1998), our hypothesis is that the increase in implied volatility represents unexpected higher default risk (in the next period) and, consequently, lower bond prices (in the future). To test our hypothesis, we investigate whether $\Delta ImpVOL$ predicts default risk of the firm. We consider various measures of default risk.

To construct the first measure, we use the procedure in Bharath and Shumway (2008) to calculate EDF . The calculation follows the insights from the Merton (1974) distance to default model:

$$EDF = N\left(-\frac{\ln(V/F) + (\mu - 0.5\sigma_V^2)T}{\sigma_V\sqrt{T}}\right), \quad (3)$$

where $N(\cdot)$ is the cumulative distribution function of the standard normal distribution, V is the total value of a firm, F is the face value of the firm's debt, μ is an estimate of the expected annual return of the firm's assets that is calculated using historical return of the firm's asset, and σ_V is the volatility of firm value. We solve for V and σ_V numerically from the following two equations:

$$E = VN(d_1) - e^{-rT}FN(d_2) \text{ and } \sigma_E = (V/E)N(d_1)\sigma_V, \quad (4)$$

where E is the market value of the firm's equity, σ_E is the volatility of the firm's equity, and d_1 and d_2 are parameters defined in the usual way. We use the code provided from Tyler Shumway's website to calculate V , σ_V , and EDF for each firm from July 2002 to August 2017.¹⁷

One concern with EDF is that it is largely driven by the ratio of the market value of the firm to the face value of the firm's debt, and the market value of the firm is, in part, driven by the market value of the firm's debt.

Because we already report a negative relation between $\Delta ImpVOL$ and future bond returns (that is, changes in the market value of the firm's debt), there could be a mechanical positive relation between $\Delta ImpVOL$ and future ΔEDF . Therefore, we need a measure that is unaffected by the market value of the firm's debt. The Campbell et al. (2008) CHS is such a measure. Accordingly, this is our second default risk measure.

The third and fourth default risk measures are at the bond level. These are changes in bond rating and a dummy variable for bond rating downgrade (equal to one if there is a future downgrade and zero otherwise). The ratings are obtained from the FISD Mergent database.

To investigate the information content of $\Delta ImpVOL$, we run Fama and MacBeth (1973) regressions for predicting future change of default risk measured over the next one, three, and six months in Table 6. The dependent variable is the change in EDF at the firm level in Panel A, change in CHS at the firm level in Panel B, rating change at the bond level in Panel C, and the dummy variable of rating downgrade at the bond level in Panel D.

The independent variables in Panels A and B include the following six market value-based accounting variables as control variables following Campbell et al. (2008): net income over market value of total assets ($NIMTA$), total liabilities over market value of total assets ($TLMTA$), logarithm of firm's market equity ($Size_equity$), stock of cash and short-term investments over the market value of total assets ($CASHMTA$), market-to-book value of the firm (MB), and price per share ($Price_equity$). We also consider three bond characteristics as control variables: the average logarithm of offering amount of all bonds in a firm ($Size_bond$), bond maturity ($Maturity$), and bond illiquidity ($Illiquidity$).

To predict bond-level default risk measures in Panels C and D, we include implied volatility ($ImpVOL$), bond maturity ($Maturity$), bond illiquidity ($Illiquidity$), bond return in the past month (Lag_bond_return), downside risk ($VaR(5\%)$), and stock return in the past month (Lag_stock_return) as control variables.

Panel A of Table 6 shows that $\Delta ImpVOL$ predicts future change of EDF . After controlling for accounting-based predictors in Campbell et al. (2008), the coefficient estimate on $\Delta ImpVOL$ is positive and statistically significant at the 5% level at all horizons. Thus, the more the increase in implied volatility, the higher the increase in EDF in the next months. If change in implied volatility predicts future lower equity returns, then our results might be a bit mechanical because equity is an important input into the EDF calculation. Note, however, that $\Delta ImpVOL$ is the average of $\Delta CVOL$ and $\Delta PVOL$. An et al. (2014) show that, whereas $\Delta PVOL$ negatively predicts future stock returns, $\Delta CVOL$ positively predicts future stock returns. We also previously remark that the predictive

Table 6. Predicting Change of Future Default Risk with $\Delta ImpVOL$

	Panel A: Change in <i>EDF</i>			Panel B: Change in <i>CHS</i>		
	One month	Three months	Six months	One month	Three months	Six months
Intercept	0.004** (2.15)	0.009* (1.93)	0.024*** (3.26)	0.011 (1.06)	0.000 (0.01)	0.007 (0.12)
$\Delta ImpVOL$	0.031*** (2.93)	0.043** (2.11)	0.082*** (3.00)	0.103*** (3.02)	0.199*** (3.85)	−0.086 (−0.83)
<i>NIMTA</i>	−0.029 (−1.00)	−0.014 (−0.19)	0.029 (0.24)	−0.809*** (−7.84)	−2.884*** (−13.92)	−6.611*** (−14.79)
<i>TLMTA</i>	−0.003 (−0.81)	−0.009 (−0.78)	−0.017 (−0.87)	−0.019** (−2.34)	−0.069*** (−3.05)	−0.146*** (−3.70)
<i>Size_equity</i>	−0.000 (−0.45)	−0.000 (−0.31)	−0.001 (−0.77)	−0.000 (−0.25)	0.004 (1.50)	0.010** (1.99)
<i>CASHMTA</i>	−0.002 (−1.04)	−0.004 (−1.00)	−0.007 (−0.99)	−0.034*** (−4.39)	−0.104*** (−5.27)	−0.189*** (−5.46)
<i>MB</i>	−0.001* (−1.75)	−0.002* (−1.74)	−0.003** (−2.29)	0.002 (1.36)	0.003 (0.71)	−0.001 (−0.14)
<i>Price_equity</i>	0.000 (1.52)	0.000 (1.62)	0.000* (1.71)	0.000*** (4.00)	0.001*** (4.35)	0.001*** (4.85)
Adjusted R^2	0.068	0.069	0.079	0.046	0.095	0.143
Observations	115,298	115,290	115,166	91,494	91,399	91,225

	Panel C: Change in <i>Rating</i>			Panel D: <i>Rating</i> downgrade		
	One month	Three months	Six months	One month	Three months	Six months
Intercept	0.014 (0.50)	0.050 (0.98)	0.076 (1.07)	0.006 (0.48)	0.020 (0.90)	0.035 (1.18)
$\Delta ImpVOL$	0.251*** (2.73)	0.386** (2.52)	0.666*** (2.86)	0.100** (2.07)	0.163** (2.35)	0.270*** (3.13)
<i>ImpVOL</i>	0.064** (2.01)	0.336*** (6.18)	0.589*** (7.35)	0.102*** (6.63)	0.219*** (8.16)	0.305*** (9.09)
<i>Size</i>	−0.004** (−2.30)	−0.007* (−1.94)	−0.009* (−1.76)	−0.002** (−2.20)	−0.004** (−2.18)	−0.005** (−2.18)
<i>Rating</i>	0.003*** (3.20)	−0.007*** (−3.85)	−0.016*** (−6.32)	0.001* (1.97)	−0.001* (−1.76)	−0.003*** (−2.75)
<i>Maturity</i>	0.000 (1.18)	−0.000 (−0.24)	−0.000 (−0.93)	0.000 (1.01)	−0.000 (−0.19)	−0.000 (−0.22)
<i>Illiquidity</i>	−0.002 (−1.51)	−0.005** (−2.28)	−0.007** (−2.37)	−0.002*** (−2.63)	−0.003*** (−3.28)	−0.003** (−2.26)
<i>Lag bond return</i>	−0.583*** (−7.43)	−0.894*** (−7.39)	−1.083*** (−6.31)	−0.232*** (−6.47)	−0.339*** (−6.16)	−0.411*** (−6.21)
<i>VaR (5%)</i>	−0.112 (−1.20)	−0.390** (−2.55)	−0.703*** (−3.66)	−0.079* (−1.90)	−0.182*** (−2.71)	−0.263*** (−3.34)
<i>Lag stock return</i>	−0.015 (−0.39)	−0.184*** (−3.07)	−0.356*** (−4.50)	−0.011 (−0.75)	−0.073*** (−2.93)	−0.128*** (−4.29)
Adjusted R^2	0.057	0.081	0.095	0.061	0.092	0.105
Observations	544,142	540,377	522,990	544,142	540,377	522,990

Notes. This table reports Fama and MacBeth (1973) cross-sectional regression results for predicting changes of future default risk with $\Delta ImpVOL$. The dependent variable is the change in expected default frequency (*EDF*) in Panel A, change in Campbell et al. (2008) (*CHS*) in Panel B, change in bond rating in Panel C, and bond rating downgrade dummy in Panel D. The dummy variable is equal to one if there is a downgrade of the bond and zero otherwise. We run regressions at the firm level in Panels A and B, and at the bond level in Panels C and D. We report regression results for changes in default risk in the future one, three, and six months in both panels. Independent variables in Panels A and B include changes in implied volatilities, net income over market value of total assets (*NIMTA*), total liabilities over market value of total assets (*TLMTA*), logarithm of firm's market equity (*Size_equity*), stock of cash and short-term investments over the market value of total assets (*CASHMTA*), market-to-book value of the firm (*MB*), price per share (*Price_equity*), and lagged *EDF* (*Lag EDF*). Independent variables in Panels C and D are the same as those in specification (2) of Table 5. Newey–West adjusted *t*-statistics are provided in parentheses. All independent variables are winsorized each month at the 0.5% level. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

power of $\Delta ImpVOL$ is weak for future stock returns. Nevertheless, as an additional test, we use only $\Delta CVOL$ in regressions in Panel A of Table 6. We find that the coefficient estimate on $\Delta CVOL$ is positive at

all horizons and statistically significant at the 5% level at the one-month horizon. Panel B shows similar results for the change in *CHS* albeit statistically insignificant for the six-month horizon.

One plausible alternative is that $\Delta ImpVOL$ contains information about future liquidity (rather than default risk) that affects both the stock and bond and has not been incorporated into the bond market yet. For example, if investors predict that liquidity for a certain firm's securities will dry up in the next month and this low liquidity could induce high future volatility, then these investors may purchase options to express this view, but the bond market may underreact to this information. To examine this hypothesis, we run Fama and MacBeth (1973) regressions in which the dependent variable is the change in bond illiquidity (*Illiquidity*) as defined in Section 2.1. In unreported results, we find that the $\Delta ImpVOL$ has no forecasting power for next month bond illiquidity.

Panels C and D show that the $\Delta ImpVOL$ significantly predicts future bond rating changes and downgrades over the next one, three, and six months after controlling for several bond characteristics. For example, if $\Delta ImpVOL$ increases by one standard deviation, the probability of downgrade increases by 2.22% ($= 0.554 \times 0.04$) over the next month and 3.95% ($= 0.988 \times 0.04$) over the next three months. Some other variables also significantly predict the probability of future downgrades. For example, higher implied volatility and lower bond return in the previous month predict higher probability of downgrade in the next one, three, and six months.

Overall, we find that changes in implied volatilities are significant predictors for future changes in *EDF* and *CHS* and future ratings changes. The evidence suggests that informed traders with relevant default risk information prefer to trade in the option market (which allows higher leverage and higher potential for profits) before they trade in the corporate bond market. Large increases in implied volatility suggest higher default risk of the firm in the future. The corporate bond market fails to immediately fully incorporate the default risk-related information, leading to a higher yield, lower price, and low return in the future.

4.2. Predicting the Change of the Credit Default Swap Spread

Another class of securities that shares many features of bonds is CDS. The price of a credit default swap is usually referred to as its "spread." If default risk of a firm increases, bond prices decrease to reflect the higher yield, and CDS prices (spreads) increase to reflect the higher price of insurance. Because our previous section shows that $\Delta ImpVOL$ predicts change in default probability, in this section, we directly study whether $\Delta ImpVOL$ predicts the change of CDS spread in a manner similar to that for the bond returns.

We obtain CDS spreads from the Markit Group Limited for a sample period of February 2001 to September 2014 (our CDS sample is close to that of Lee

et al. 2018, Augustin and Izhakian 2019). The data set contains daily quotes on CDS spreads for more than 1,000 North American firms. The average number of observations per month in our sample period is 1,004. We focus on the five-year CDS spreads because these contracts are the most liquid and constitute more than 85% of the entire CDS market. To maintain uniformity in contracts and compatibility with previous studies, such as in Griffin et al. (2016) and Lee et al. (2018), we only keep CDS for senior unsecured debt with a modified restructuring clause and denominated in U.S. dollars. To measure changes in CDS spreads, Ericsson et al. (2009) use level change, whereas Hilscher et al. (2015) and Lee et al. (2018) rely on the percentage change of CDS spread. In our work, we use both the monthly percentage change as well as the level change of CDS spread.

Table 7 reports the time-series averages of Fama and MacBeth (1973) regression coefficients and their corresponding *t*-statistics. We run firm-level predictive regressions at monthly frequency with the change of CDS spread as the dependent variable. Independent variables include $\Delta ImpVOL$, the percentage change of CDS spreads over the last month, stock return in the past month, bond return in the past month, and implied volatility level. Bond return at the firm level is calculated as the value-weighted return of all bonds of each firm. We find that $\Delta ImpVOL$ significantly predicts future CDS spread changes after controlling for lagged CDS spread change and stock and bond returns. In addition, we find lagged returns of related securities also significantly predict future changes in CDS spread. Firms with larger lagged CDS spread change, larger lagged stock return, and larger lagged bond return tend to experience a higher CDS spread change in the next month. Results are similar using level change of CDS spread versus percentage change of CDS spread.¹⁸

To sum up, we find that $\Delta ImpVOL$ leads the CDS market similarly to that for the bond market. The evidence is consistent with the explanation that an increase in implied volatility predicts higher default risk of the underlying firm, leading to a lower bond return and an increase of CDS price. Hilscher et al. (2015) find that stock return leads the CDS market, whereas Lee et al. (2018) argue that CDS spread contributes to the price discovery in both the stock and bond markets. Our study adds to this line of research by showing that the option market contains information that is useful to predict CDS spread changes.

5. Why Does $\Delta ImpVOL$ Predict Future Bond Returns?

Why do the corporate bonds fail to immediately impound the relevant information about the change

Table 7. Predicting CDS Spread Change with $\Delta ImpVOL$

	Percentage change of CDS spread		Level change of CDS spread	
<i>Intercept</i>	0.011 (0.92)	0.013 (0.93)	0.000 (0.26)	−0.000 (−0.38)
<i>$\Delta ImpVOL$</i>	0.476** (2.48)	0.390** (2.09)	0.021*** (3.98)	0.011** (2.28)
<i>Lag % change of CDS spread</i>	−0.030** (−2.13)	−0.038*** (−2.79)	−0.000 (−0.69)	−0.001 (−1.26)
<i>Lag stock return</i>		−0.400*** (−14.34)		−0.020*** (−9.13)
<i>Lag bond return</i>		−0.213*** (−3.64)		−0.020*** (−3.98)
<i>ImpVOL</i>		−0.004 (−0.22)		0.002 (0.93)
Adjusted R^2	0.025	0.087	0.043	0.208
Observations	57,697	57,697	57,697	57,697

Notes. We run firm-level predictive regressions at monthly frequency with the percentage or level change of CDS spread as the dependent variable. Independent variables include change in implied volatility, percentage change of CDS spread over previous month, stock return in the past month, bond return in the past month, and implied volatility. Bond return at the firm level is calculated as value-weighted return of all bonds of each firm. The table reports time-series averages of the Fama and MacBeth (1973) regression coefficients and their corresponding Newey–West adjusted t -statistics. All variables are winsorized each month at the 0.5% level. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from August 2002 to August 2017.

in firm risk into bond prices? The underreaction in corporate bond price might reflect slow diffusion of information from options to bonds or impediments to trade in the corporate bond market. In this section, we provide additional evidence to bear on these two hypotheses. First, we study the effect of bond and option trading volume on bond return predictability. Second, we investigate the role of firm dual ownership as an investor-attention measure in explaining the bond return predictability. Finally, we examine how transaction cost affects the magnitude of the predictability.

We acknowledge up front that it is not possible to completely disentangle the two hypotheses. For example, the speed of information incorporation might depend on the liquidity of the corporate bond market—the lower the bond market liquidity, the slower the information reflected in the bond price. Our modest goal is only to present evidence supporting one or the other hypothesis.

5.1. Option Volume, Bond Volume, and Informed Trading

Easley et al. (1998) construct a sequential trading model to understand informed trading in the option and stock markets. They show that, if at least some informed investors choose to trade in options before trading in underlying stocks, option prices predict future stock price movements. This intuition can be echoed to the informational role of options in the bond market. If the informed trading hypothesis is correct, we expect the predictive power of $\Delta ImpVOL$ to be stronger when more informed traders choose to trade in the option market and fewer informed traders

trade in the bond market. We, therefore, analyze portfolios sorted on $\Delta ImpVOL$ conditional on changes in option and bond trading volumes.

Each month, we first divide the bonds into two separate groups based on the median change in option or bond trading volume. For example, bonds with above (below) median change in option trading volume are in the high (low) $\Delta option$ volume group. Similarly, bonds with above (below) median change in bond trading volume are in the high (low) $\Delta bond$ volume group. For each of these four groups, we further sort the bonds by $\Delta ImpVOL$ into 10 deciles and hold the portfolio for one month. We report the mean returns of the decile portfolios, the 10–1 return, and alpha from the bond + stock 11-factor model in Table 8.

Consistent with the sequential trading model, we find that the predictability of $\Delta ImpVOL$ is the strongest for bonds in the high $\Delta option$ volume and low $\Delta bond$ volume groups. The average return and bond + stock alpha for the 10–1 portfolio are −1.02% and −1.29% with t -statistics of −3.26 and −3.80, respectively, for this subsample. The risk-adjusted return spreads are smallest in the group of low $\Delta option$ volume and low $\Delta bond$ volume at −0.72% per month.

Note also that, whereas there is a variation in profits across the four groups, we continue to find statistically and economically significant profits in all groups. For instance, even in the groups of high $\Delta option$ volume and high $\Delta bond$ volume, in which we expect the least limits to arbitrage, we find that the risk-adjusted return spreads are −0.82% per month.

We acknowledge that one of the presumptions behind our test in this section is that changes in trading volumes measure where informed traders are

Table 8. Returns and Alphas of Portfolio of Bonds Sorted by $\Delta ImpVOL$ Conditional on Changes in Option and Bond Trading Volume

$\Delta Option$ volume $\Delta Bond$ volume	High Low	High High	Low Low	Low High
1	1.03	1.03	0.88	0.94
2	0.72	0.67	0.76	0.64
3	0.64	0.60	0.68	0.53
4	0.53	0.72	0.65	0.50
5	0.54	0.59	0.50	0.51
6	0.42	0.43	0.53	0.49
7	0.30	0.44	0.29	0.50
8	0.62	0.59	0.45	0.57
9	0.39	0.35	0.16	0.36
10	0.01	0.16	0.62	0.27
10 – 1	–1.02***	–0.88***	–0.27*	–0.67***
	(–3.26)	(–2.78)	(–1.90)	(–3.24)
Bond + stock alpha	–1.29***	–0.82***	–0.72***	–0.92***
	(–3.80)	(–3.51)	(–3.16)	(–4.64)

Notes. This table reports portfolio returns sorted by $\Delta ImpVOL$ ($\equiv \Delta CVOL + \Delta PVOL$) conditional on changes in option and bond trading volume. For each month, we separate the bonds into two groups based on the median change in option trading volume ($\Delta Option$ volume) or median change in bond trading volume ($\Delta Bond$ volume). For bonds in the intersection of each of these four groups, we further sort the bonds by $\Delta ImpVOL$. We report the mean return of the 10 portfolios and the 10–1 return in this table. Alpha is calculated from the bond + stock 11-factor model. Newey–West adjusted t -statistics are in parentheses. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

trading. However, it is an empirical question whether levels or changes in volume proxy for informed trading. Therefore, we run similar tests as in Table 8, but instead condition on levels of option and bond volume. In unreported results, we find again that predictability of $\Delta ImpVOL$ is the strongest for bonds in the high option volume and low bond volume groups.¹⁹ The average return and bond + stock alpha for the 10–1 portfolio are –1.06% and –1.31% with t -statistics of –2.49 and –6.05, respectively, for this subsample.

5.2. Investor Inattention and Bond Return Predictability

The speed at which asset prices incorporate new information is affected by investors' limited attention. Limited attention can cause investors to ignore useful information, leading to price underreaction. Theoretical models, such as Hirshleifer and Teoh (2003) and Hirshleifer et al. (2011) show how limited attention causes underreactions to news. There are numerous empirical studies on the effects of investors' limited attention (see, for example, Peng and Xiong 2006, Barber and Odean 2008, DellaVigna and Pollet 2009). In this section, we use rating change announcements and institutional dual ownership of a firm's stock and bond to examine the role of investors' attention on the predictability of $\Delta ImpVOL$ on bond returns.

5.2.1. Ratings Announcement Days. Engelberg et al. (2018) show that anomaly returns in the stock market are 50% higher on corporate news days and six times higher on earnings announcement days. This is consistent with the idea that biased expectations drive anomaly returns, and they are partially corrected upon news arrival. Following this insight, we conjecture that the long–short bond portfolio return sorted by $\Delta ImpVOL$ is higher in magnitude around bond rating announcements, which are the most salient events in the corporate bond market.

To examine the effect of rating announcements on our anomaly returns, we follow Engelberg et al. (2018) and first run a panel regression of daily bond return in deciles 1 and 10 on the rating day dummy variable. The dependent variable is the daily bond return multiplied by 100. For each bond–month observation, we define a *Net* variable, which is equal to –1 if the bond belongs to decile 1 and equal to 1 if the bond belongs to decile 10. The rating day indicator (*RDAY*) equals one if the day is in one of the three-day windows around a rating announcement and equals zero on other days. We also include an interaction term, *Net* $\times$ *RDAY*, which indicates whether anomaly returns are higher or lower on rating days. Our panel regression includes time fixed effects, and we cluster standard errors at the time level. We report the panel regression results in Panel A of Table 9. We find that the coefficient on *Net* is –0.02 (t -statistic = –6.52), whereas that on *Net* $\times$ *RDAY* is –0.11 (t -statistic = –2.87). Thus, the long–short return is 5.5 times higher in magnitude on rating announcements than on other days.

We report average daily returns (in bps) of deciles 1 and 10 on rating days and other days in Panel B of Table 9. We further split the rating announcements into downgrades and upgrades. We find that the long–short return is 8.3 times higher in magnitude on rating days (–25 bps) than other days (–3 bps). For downgrade announcement days, the anomaly return is as large as –58 bps. For upgrades, the anomaly return is 0.27 bps. The results are consistent with the idea that investors lower (raise) their expectations around downgrade (upgrade) announcements. The positive return of 17 bps on decile 10 around upgrades spotlights events in which our signal of increase in implied volatility (implying increased risk and lower bond prices) is false. Correspondingly, our signal of an increase in implied volatility is deemed to be materially important for decile 10 on downgrades when the bond return is –82 bps.

Finally, we check the effects of earnings announcement days on bond returns. Atilgan (2014) finds that predictability of optionable stock returns is more pronounced during a two-day earnings announcement window. We eliminate all earnings announcement months from our sample, which reduces the sample

Table 9. Bond Return Predictability Around Rating Announcement Days

Panel A: Panel regression of bond returns on rating announcement days				
	<i>Net</i>	<i>Net</i> × <i>RDAY</i>	<i>RDAY</i>	
	−0.02*** (−6.52)	−0.11*** (−2.87)	−0.32*** (−8.47)	
Panel B: Returns of deciles 1 and 10 on rating days and other days				
	Rating day	Downgrade	Upgrade	Other days
Decile 1	−0.15	−0.24	−0.10	0.06
Decile 10	−0.40	−0.82	0.17	0.03
10−1	−0.25	−0.58	0.27	−0.03

Notes. This table reports bond returns around rating announcement days. Panel A reports results from a panel regression of daily returns on time fixed effects, the *Net* anomaly variable, rating-day dummy variable *RDAY*, and interaction between *Net* and *RDAY*. The dependent variable, daily bond return, is multiplied by 100. For each bond-month observation, the *Net* variable equals −1 if the bond belongs to decile 1 and equals 1 if the bond belongs to decile 10. *RDAY* is a dummy variable that equals one if the day is in one of the three days around a rating announcement and zero otherwise. We report *t*-statistics in parenthesis using standard errors clustered on time. Panel B reports average daily returns of deciles 1 and 10 on rating days and other days. We further split the rating announcements into downgrades and upgrades. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

size by one third. Nevertheless, even in this reduced sample, the stock + bond alpha of the 10–1 spread is −0.74% (*t*-statistic = −4.12), which is close to our baseline results of −0.98%. We conclude that earnings announcement days are not salient events for our predictability results.

5.2.2. Dual Ownership. Investors' attention can also be captured by institutional dual ownership of a firm's stock and bond. Following Bodnaruk and Rossi (2016), we define dual institutions for a company as financial institutions that hold at least 0.5% of stock and 0.5% of bond of that company. We then calculate the dual institutional ownership at the bond level by aggregating the ownership of all dual institutions. Firms without dual institutions are defined to have zero dual ownership.²⁰ High dual ownership bonds are defined as bonds with dual institutional ownership above the median. Low dual ownership bonds are bonds with dual institutional ownership below the median. The dual ownership variable is available at an annual frequency from 2006 to 2015. The sample is relatively large-sized firms in the S&P 1500 index. Intuitively, financial institutions that hold both equity and bond of the same firm at the same time pay more attention to the information related to stock options than those that hold only bonds of the firm. Thus, we expect the predictability of $\Delta ImpVOL$ on future bond returns to be stronger for firms without dual ownership.

We report portfolio sorting results for the full subsample of S&P 1500 firms, firms with low dual ownership, and firms with high dual ownership in the sample period of 2007 to 2016 in Table 10. The results for the sample of S&P 1500 firms are similar to those in Table 2. This illustrates that our sample selection does not create a bias for main results. Consistent with the

investor inattention hypothesis, we find the portfolio return spread is larger in magnitude for the bonds with low dual ownership firms at −0.83% (*t*-statistic = −2.06) than that for bonds with high dual ownership firms at −0.49% (*t*-statistic = −2.10).

The results in this section, thus, suggest that investor attention plays a role in explaining the slow information diffusion from the option market to the corporate bond market.

5.3. Transaction Cost Analysis

Recall that the results in Panel A of Table 4 show that predictability exists in all bonds regardless of their illiquidity and also shows that the predictability is highest among more illiquid bonds. Section 5.1 also shows

Table 10. Impact of Firm-Level Dual Ownership on Bond Return Predictability

	1	2	3	4	5	5–1
S&P 1500 sample	0.82 (−2.71)	0.70 (−3.38)	0.50 (−2.95)	0.43 (−2.76)	0.20 (−0.84)	−0.61** (−2.31)
Low dual ownership	0.89 (−2.73)	0.66 (−2.78)	0.58 (−2.87)	0.4 (−2.38)	0.06 (−0.18)	−0.83** (−2.06)
High dual ownership	0.76 (−2.44)	0.71 (−3.86)	0.42 (−2.58)	0.45 (−2.74)	0.27 (−1.46)	−0.49** (−2.10)

Notes. We report returns (and Newey–West adjusted *t*-statistics in parentheses) of portfolio for bonds with low and high dual ownership in the S&P 1500 sample. We first define dual institution for a company as financial institutions that hold at least 0.5% of the outstanding stocks and 0.5% of the outstanding bonds of that company. Then, we calculate the dual institutional ownership at the bond level by aggregating the ownership of all dual institutions. High dual ownership bonds are defined as bonds with dual institutional ownership above the median. Low and zero dual ownership bonds are bond with dual institutional ownership below the median. The dual ownership is calculated on an annual basis. For each group, bonds are sorted every month in quintiles by $\Delta ImpVOL$. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from January 2007 to December 2016.

that our strategy is profitable in even illiquid bonds. However, both these analyses do not explicitly account for the impact of illiquidity in trading. This is because, so far, we have assumed that all bonds are bought or sold at the volume-weighted transaction price at the month end.

To examine the impact of trading cost on the profitability of our strategy, we estimate transaction costs using two approaches. In the first approach, we use the mean bid–ask spread estimates from Edwards et al. (2007). The relevant trading costs, EHP, are 18, 16, and 30 bps (68, 45, and 100 bps) for all bonds, investment-grade bonds, and non-investment-grade bonds, respectively, for trade size of \$1M (\$100K). In the second approach, we use the Bao et al. (2011) measure (BPW). This is calculated as $2\sqrt{\gamma}$, where γ is the illiquidity measure in Roll (1984):

$$\gamma = \begin{cases} -\text{cov}(r_d, r_{d-1}) & \text{if } \text{cov}(r_d, r_{d-1}) < 0 \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where r_d is the corporate bond return on day d .

We report portfolio turnover, bid–ask spread, net return, and net alphas from the bond and bond + stock factor model for the long–short portfolio sorted by ΔImpVOL in Table 11. The table shows results for the full sample and for the subsamples of investment-grade, and non-investment-grade bonds. Turnover is defined as the average sum of the percentage of a portfolio that is bought and the percentage of a portfolio that is sold in each month. Bid–ask spread is estimated using the EHP estimates or the BPW measure. Net return is the portfolio return net of transaction costs. The factor models are the same as those in Table 2.

For consistency with earlier tables, the hedge portfolio is defined to be long in decile 10 and short in

decile 1. Thus, negative net returns show a profitable strategy, whereas positive net returns show the lack of profitability accounting for transaction costs. Table 11 shows that net return of the long–short portfolio in the full sample is statistically significant if we use the EHP estimates for a trade size of \$1M. The alphas from the factor models are also statistically significant. When we use EHP estimates for a trade size of \$100K, net return and alphas are no longer statistically significant and even reverse sign. Because the Bao et al. (2011) trading costs are much higher than those of Edwards et al. (2007), the trading strategy does not survive transaction costs using BPW bid–ask spreads—net returns for investment-grade and non-investment-grade samples are not statistically significant after subtracting the transaction cost. Overall, our results show that the trading strategy could be potentially profitable for large trading sizes. Nevertheless, high transaction costs might hinder trades that seek to exploit this arbitrage opportunity.²¹

To summarize the results in this section, we find that slow diffusion of information is largely responsible for predictability from options to bonds. Whereas predictability is the highest when option trading volumes are high and when bond trading volumes are low, there is predictability in even very liquid bonds. Predictability is muted for bonds of firms in which investors pay attention to both stocks and options. This lack of investor attention creates opportunities for arbitrageurs. However, the transaction cost analysis highlights the difficulty in taking advantage of this predictability. Thus, slow diffusion of information because of investor inattention coupled with high limits to arbitrage explains why bond prices underreact to information from options.

Table 11. Transaction Costs and Net Returns for Long–Short Portfolio Sorted by ΔImpVOL

	EHP (\$1M)			EHP (\$100K)			BPW		
	All	IG	Junk	All	IG	Junk	All	IG	Junk
Turnover	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71
Bid–ask spread	0.18	0.16	0.30	0.68	0.45	1.00	1.29	1.19	1.68
Net return	−0.30*	−0.04	−0.32	0.53	0.82	0.53	1.51	1.46	2.30
	(−1.74)	(−0.31)	(−1.19)	(3.07)	(6.17)	(1.97)	(5.89)	(6.25)	(6.11)
Bond alpha	−0.53***	−0.39**	−0.36	0.29	0.45	0.48	0.85	0.75	1.50
	(−3.26)	(−2.43)	(−1.08)	(1.78)	(2.72)	(1.45)	(5.10)	(5.58)	(4.79)
Bond + stock alpha	−0.68***	−0.52***	−0.54*	0.14	0.32	0.29	0.77	0.68	1.41
	(−3.74)	(−2.74)	(−1.69)	(0.75)	(1.66)	(0.91)	(4.50)	(5.01)	(4.67)

Notes. This table presents portfolio results for bonds sorted by ΔImpVOL after subtracting the transaction costs. We calculate returns on a long–short portfolio that is long the 10th and short the 1st decile. We form the portfolio for the full sample and for the subsamples of investment-grade bonds (IG) and non-investment-grade bonds (Junk). We use the mean bid–ask spread estimates from Edwards et al. (2007) (EHP) for trade size \$1M and \$100K. We also calculate bid–ask spread following Bao et al. (2011) (BPW). Turnover is defined as the average sum of the percentage of a portfolio that is bought and the percentage of a portfolio that is sold in each month. Net return is the portfolio return net of transaction costs. Positive net return indicates lack of predictability after accounting for transaction costs. We also report alpha from the bond factor model and the bond + stock factor model. The factor models are the same as those in Table 2. All returns are in percentage per month. The table reports these alphas together with their Newey–West adjusted t -statistics in parenthesis. *, **, and *** represent 10%, 5%, and 1% significance levels. The sample period is from July 2002 to August 2017.

6. Conclusion

The price discovery role of options for the underlying stocks is well-documented in the literature. The price discovery role of options on the bond market, however, is unknown. In this paper, we investigate whether option information contains relevant information for the future return of the corporate bond of the same underlying firm. In particular, we study implied volatility changes in the past month when the implied volatility is obtained from at-the-money options with 365 days of maturity. We find that the firms with a large increase in implied volatilities have a low corporate bond return in the next month. If we form decile portfolios based on changes in implied volatilities, the spread in average return between the top and bottom decile portfolios is approximately 0.6% per month and highly statistically significant. The predictability of implied volatility changes on bond return is robust after controlling for stock and bond risk factors, bond characteristics, and other volatility characteristics. We further document that implied volatility changes have relevant information for predicting changes in default risk in the next months and for predicting future changes of credit default swap spreads. We also find evidence consistent with investor inattention driving predictability. The predictability is lower in low option and bond volume and in bonds with low dual ownership of stocks and bonds holding by the same financial institutions. High transaction costs, however, present an important source of limits to arbitrage, which makes the bond market slow in incorporating information from the option market.

Acknowledgments

The authors thank the editor, Haoxiang Zhu; an anonymous associate editor; and two anonymous reviewers as well as Jack Bao, Olivier Blin, Guido Bolliger, Zhi Da, Kent Daniel, Kewei Hou, Jianfeng Hu, Ron Kaniel, Neil Pearson, M. Fabricio Perez, Luis Goncalves-Pinto, Maureen O'Hara, Norman Seeger, and Sheridan Titman as well as seminar participants at Bayes Business School, Chinese University of Hong Kong, Indian School of Business Hyderabad, Singapore Management University, Sun Yat-sen University, Unigestion, and Vrije University Amsterdam for helpful discussions and useful suggestions. The authors have benefited from the comments of participants at the 2019 Northern Finance Association Annual Meeting, 2019 OptionMetrics Conference, 2019 Taiwan Finance Association Conference, the Second Derivatives and Quantitative Investing Conference at Chinese University of Hong Kong, the 2020 European Finance Association Annual Meeting, and the 2021 American Finance Association Annual Meeting. All errors are their own.

Endnotes

¹ See, for example, An et al. (2014), Bali and Hovakimian (2009), Chakravarty et al. (2004), Conrad et al. (2013), Cremers and Weinbaum (2010), Johnson and So (2012), Pan and Poteshman (2006), Stilger et al. (2017), and Xing et al. (2010).

² Campbell and Taksler (2003) find that equity volatility explains cross-sectional variation in bond yields. Shumway (2001) uses equity volatility as one of the inputs in his bankruptcy prediction model. Christensen and Prabhala (1998) show that implied volatility is a good predictor of future volatility.

³ Relatedly, Du et al. (2019) find that stochastic volatility creates a wedge between debt and equity. As such, changes in volatility could presumably have different effects on different claims on assets of a firm.

⁴ There are some studies that use information from options to predict CDS prices (spreads) or returns for a small sample of firms. For example, Blanco et al. (2005) study 33 firms, and Cao et al. (2010) study a sample of 301 firms.

⁵ See also Lin et al. (2018), who analyze a large set of predictors to predict aggregate bond returns.

⁶ Factor investing papers construct bond factors using information from bonds (for example, Gebhardt et al. 2005a, Jostova et al. 2013, Houweling and van Zundert 2017), from stocks (for example, Chordia et al. 2017, Bektić et al. 2019), or from stocks and bonds (for example, Israel et al. 2018).

⁷ We thank Jens Dick-Nielsen for providing the SAS program to clean the reporting errors from the Enhanced TRACE data set.

⁸ Our main findings are not sensitive to imposing this filter.

⁹ Bond rating is the average of ratings provided by S&P and Moody's when both are available or the rating provided by one of the two rating agencies when only one rating is available.

¹⁰ OptionMetrics records option prices (used in implied volatility calculations) up to March 4, 2008, as end-of-day prices (i.e., 4:15 p.m. EST). This means there is look-ahead bias in the option prices and implied volatilities because they reflect 15 minutes of extra price variation. Our use of a skipping day avoids this issue. We also find that results are, in fact, stronger in the post-2008 sample period (during which OptionMetrics records prices as close as possible to 4:00 p.m. EST).

¹¹ We thank Turan Bali for providing us data on these factors and refer the readers to Bai et al. (2019) for details on the construction of these factors.

¹² We also calculate alphas from a factor model with lagged factors in addition to contemporaneous factors. This makes very little quantitative difference to the magnitude of alphas. For instance, the alpha from a bond + stock factor model is still -0.84% (t -statistic = -4.67). Finally, we also calculate alphas and betas from the bond three-factor model of Fama and French (1993). This model includes a market bond factor, a term factor, and a default factor. We find that the betas of the long-short portfolio on these factors are small and statistically insignificantly different from zero. Moreover, the premia on the term and the default factor are close to zero, which partially justifies the new state-of-the-art factor model of Bai et al. (2019).

¹³ We also check stock portfolio returns. Similar to An et al. (2014), we find that sorts on $\Delta CVOL - \Delta PVOL$ produce a bond + stock alpha of a 10–1 stock portfolio of 0.82% per month. However, sorts on $\Delta CVOL + \Delta PVOL$ produce a bond + stock alpha of the 10–1 stock portfolio of -0.53% per month albeit statistically insignificant. At the same time, the formation-month bond + stock alpha of the 10–1 stock portfolio for sorts on $\Delta CVOL + \Delta PVOL$ is -9.93% per month, reminiscent of Black's (1976) leverage effect.

¹⁴ We also do sorts using percentage change in $ImpVOL$ rather than simple differences. In unreported results, we find that 10–1 spread in bond portfolio returns for $\% \Delta ImpVOL$ -sorted portfolios is slightly lower at -0.40% (t -statistic = -3.93).

¹⁵ Bao and Pan (2013) report that the volatility from daily returns in corporate bonds may reflect bid–ask bounce and, therefore, could

be just another illiquidity proxy. Accordingly, we also calculate bond volatility using monthly returns in the past 36 months and find that it has no material effect on our results in both this section as well as the next section on Fama–MacBeth regressions.

¹⁶ We also try other measures of uncertainty, such as analyst forecast dispersion. However, we find that neither the forecast dispersion nor its change has any predictive power for bond returns.

¹⁷ We thank Tyler Shumway for providing his SAS code on his website. See http://www-personal.umich.edu/~shumway/papers.dir/nuiter99_print.sas.

¹⁸ Cao et al. (2010) find that put implied volatility predicts CDS spreads. Our paper focuses on changes in CDS spreads. Nevertheless, we do find that changes in put-implied volatility predicts changes in CDS spreads.

¹⁹ Cao et al. (2021) find that firms with actively traded options shift from bank loan financing to public debt financing. Our study provides a plausible channel: the publicly available option prices contain default-related information in particular when option trading volumes are higher. It, to some extent, could alleviate the information disadvantage of public bond investors when compared with the banks.

²⁰ We thank Tao Chen for sharing the data set on the dual ownership variable. The details of data construction can be found in Chen et al. (2020).

²¹ It is well-documented that trading costs in CDS markets are substantially lower than those in the bond market. For example, Biswas et al. (2015) show that, for trade sizes of \$500K, bonds are three times more expensive to trade than CDS written on them. Therefore, investors could potentially explore the profitability through corporate bond market or CDS market depending on the transaction costs and trade size.

References

- An B-J, Ang A, Bali TG, Cakici N (2014) The joint cross section of stocks and options. *J. Finance* 69(5):2279–2337.
- Ang A, Hodrick RJ, Xing Y, Zhang X (2006) The cross section of volatility and expected returns. *J. Finance* 61(1):259–299.
- Atilgan Y (2014) Volatility spreads and earnings announcement returns. *J. Banking Finance* 38:205–215.
- Augustin P, Izhakian Y (2019) Ambiguity, volatility, and credit risk. *Rev. Financial Stud.* 33(4):1618–1672.
- Back K (1993) Asymmetric information and options. *Rev. Financial Stud.* 6(3):435–472.
- Bai J, Bali TG, Wen Q (2019) Common risk factors in the cross-section of corporate bond returns. *J. Financial Econ.* 131(3):619–642.
- Bali TG, Hovakimian A (2009) Volatility spreads and expected stock returns. *Management Sci.* 55(11):1797–1812.
- Bali TG, Subrahmanyam A, Wen Q (2021) Long-term reversals in the corporate bond market. *J. Financial Econ.* 139(2):656–677.
- Bao J, Pan J (2013) Bond illiquidity and excess volatility. *Rev. Financial Stud.* 26(12):3068–3103.
- Bao J, Pan J, Wang J (2011) The illiquidity of corporate bonds. *J. Finance* 66(3):911–946.
- Barber BM, Odean T (2008) All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Rev. Financial Stud.* 21(2):785–818.
- Bektić D, Wenzler J-S, Wegener M, Schiereck D, Spielmann T (2019) Extending Fama–French factors to corporate bond markets. *J. Portfolio Management* 45(3):141–158.
- Bessembinder H, Maxwell WF, Venkataraman K (2006) Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *J. Financial Econ.* 82(2):251–288.
- Bessembinder H, Kahle KM, Maxwell WF, Xu D (2009) Measuring abnormal bond performance. *Rev. Financial Stud.* 22(10):4219–4258.
- Bharath ST, Shumway T (2008) Forecasting default with the merton distance to default model. *Rev. Financial Stud.* 21(3):1339–1369.
- Biais B, Hillion P (1994) Insider and liquidity trading in stock and options markets. *Rev. Financial Stud.* 7(4):743–780.
- Biswas G, Nikolova S, Stahel CW (2015) The transaction costs of trading corporate credit. Preprint, submitted December 2, 2014, <https://dx.doi.org/10.2139/ssrn.2532805>.
- Black F (1976) Studies of stock price volatility changes. *Proc. Bus. Econom. Section Amer. Statist. Assoc.*, 177–181.
- Black F, Scholes M (1973) The pricing of options and corporate liabilities. *J. Political Econ.* 81(3):637–654.
- Blanco R, Brennan S, Marsh IW (2005) An empirical analysis of the dynamic relation between investment-grade bonds and credit default swaps. *J. Finance* 60(5):2255–2281.
- Bodnaruk A, Rossi M (2016) Dual ownership, returns, and voting in mergers. *J. Financial Econ.* 120(1):58–80.
- Bongaerts D, de Jong F, Driessen J (2017) An asset pricing approach to liquidity effects in corporate bond markets. *Rev. Financial Stud.* 30(4):1229–1269.
- Campbell JY, Taksler GB (2003) Equity volatility and corporate bond yields. *J. Finance* 58(6):2321–2349.
- Campbell JY, Hilscher J, Szilagyi J (2008) In search of distress risk. *J. Finance* 63(6):2899–2939.
- Cao C, Yu F, Zhong Z (2010) The information content of option-implied volatility for credit default swap valuation. *J. Financial Markets* 13(3):321–343.
- Cao J, Hertzel M, Xu J, Zhan X (2021) Options trading and corporate debt structure. Preprint, submitted February 10, 2020, <https://dx.doi.org/10.2139/ssrn.3520403>.
- Chakravarty S, Gulen H, Mayhew S (2004) Informed trading in stock and option markets. *J. Finance* 59(3):1235–1257.
- Chava S, Jarrow RA (2004) Bankruptcy prediction with industry effects. *Rev. Finance* 8(4):537–569.
- Chen T, Zhang L, Zhu Q (2020) Dual ownership and risk-taking incentives in managerial compensation. Working paper, Nanyang Technological University, Singapore.
- Choi J, Kim Y (2018) Anomalies and market (dis)integration. *J. Monetary Econ.* 100:16–34.
- Choi J, Richardson M (2016) The volatility of a firm's assets and the leverage effect. *J. Financial Econ.* 121(2):254–277.
- Chordia T, Goyal A, Nozawa Y, Subrahmanyam A, Tong Q (2017) Are capital market anomalies common to equity and corporate bond markets? An empirical investigation. *J. Financial Quant. Anal.* 52(4):1301–1342.
- Christensen BJ, Prabhala NR (1998) The relation between implied and realized volatility. *J. Financial Econ.* 50(2):125–150.
- Chung KH, Wang J, Wu C (2019) Volatility and the cross-section of corporate bond returns. *J. Financial Econ.* 133(2):397–417.
- Conrad J, Dittmar RF, Ghysels E (2013) Ex ante skewness and expected stock returns. *J. Finance* 68(1):85–124.
- Cremers M, Weinbaum D (2010) Deviations from put-call parity and stock return predictability. *J. Financial Quant. Anal.* 45(2):335–367.
- Cremers KJM, Driessen J, Maenhout P (2008) Explaining the level of credit spreads: Option-implied jump risk premia in a firm value model. *Rev. Financial Stud.* 21(5):2209–2242.
- DellaVigna S, Pollet JM (2009) Investor inattention and Friday earnings announcements. *J. Finance* 64(2):709–749.
- Du D, Elkahmi R, Ericsson J (2019) Time-varying asset volatility and the credit spread puzzle. *J. Finance* 74(4):1841–1885.
- Easley D, O'Hara M, Srinivas PS (1998) Option volume and stock prices: Evidence on where informed traders trade. *J. Finance* 53(2):431–465.
- Edwards AK, Harris LE, Piwowar MS (2007) Corporate bond market transaction costs and transparency. *J. Finance* 62(3):1421–1451.
- Engelberg J, McLean RD, Pontiff J (2018) Anomalies and news. *J. Finance* 73(5):1971–2001.

- Ericsson J, Jacobs K, Oviedo R (2009) The determinants of credit default swap premia. *J. Financial Quant. Anal.* 44(1):109–132.
- Fama EF, French KR (1993) Common risk factors in the returns on stocks and bonds. *J. Financial Econom.* 33(1):3–56.
- Fama EF, French KR (2015) A five-factor asset pricing model. *J. Financial Econom.* 116(1):1–22.
- Fama EF, French KR (2018) Choosing factors. *J. Financial Econom.* 128(2):234–252.
- Fama EF, MacBeth JD (1973) Risk and return: Some empirical tests. *J. Political Econom.* 81(3):607–636.
- Figlewski S, Webb GP (1993) Options, short sales, and market completeness. *J. Finance* 48(2):761–777.
- Gebhardt WR, Hvidkjaer S, Swaminathan B (2005a) The cross-section of expected corporate bond returns: Betas or characteristics? *J. Financial Econom.* 75(1):85–114.
- Gebhardt WR, Hvidkjaer S, Swaminathan B (2005b) Stock and bond market interaction: Does momentum spill over? *J. Financial Econom.* 75(3):651–690.
- Gervais S, Kaniel R, Mingelgrin DH (2001) The high-volume return premium. *J. Finance* 56(3):877–919.
- Goncalves-Pinto L, Grundy BD, Hameed A, van der Heijden T, Zhu Y (2020) Why do option prices predict stock returns? The role of price pressure in the stock market. *Management Sci.* 66(9):3903–3926.
- Griffin PA, Hong HA, Kim J-B (2016) Price discovery in the CDS market: The informational role of equity short interest. *Rev. Accounting Stud.* 21(4):1116–1148.
- Hilscher J, Pollet JM, Wilson M (2015) Are credit default swaps a sideshow? Evidence that information flows from equity to CDS markets. *J. Financial Quant. Anal.* 50(3):543–567.
- Hirshleifer D, Teoh SH (2003) Limited attention, information disclosure, and financial reporting. *J. Accounting Econom.* 36(1–3):337–386.
- Hirshleifer D, Lim SS, Teoh SH (2011) Limited investor attention and stock market misreactions to accounting information. *Rev. Asset Pricing Stud.* 1(1):35–73.
- Hong H, Sraer D (2013) Quiet bubbles. *J. Financial Econom.* 110(3):596–606.
- Houweling P, van Zundert J (2017) Factor investing in the corporate bond market. *Financial Anal. J.* 73(2):100–115.
- Hu J (2014) Does option trading convey stock price information? *J. Financial Econom.* 111(3):625–645.
- Israel R, Palhares D, Richardson S (2018) Common factors in corporate bond returns. *J. Investment Management* 16:17–46.
- Johnson TL, So EC (2012) The option to stock volume ratio and future returns. *J. Financial Econom.* 106(2):262–286.
- Jostova G, Nikolova S, Philipov A, Stahel CW (2013) Momentum in corporate bond returns. *Rev. Financial Stud.* 26(7):1649–1693.
- Lee J, Naranjo A, Velioglu G (2018) When do CDS spreads lead? Rating events, private entities, and firm-specific information flows. *J. Financial Econom.* 130(3):556–578.
- Lin H, Junbo W, Chunchi W (2011) Liquidity risk and expected corporate bond returns. *J. Financial Econom.* 99(3):628–650.
- Lin H, Wu C, Zhou G (2018) Forecasting corporate bond returns with a large set of predictors: An iterated combination approach. *Management Sci.* 64(9):4218–4238.
- Kwan SH (1996) Firm-specific information and the correlation between individual stocks and bonds. *J. Financial Econom.* 40(1):63–80.
- Merton RC (1973) Theory of rational option pricing. *Bell J. Econom.* 4(1):141–183.
- Merton RC (1974) On the pricing of corporate debt: The risk structure of interest rates. *J. Finance* 29(2):449–470.
- Navon Y (2014) The joint cross section of options and bonds. Preprint, submitted December 22, <https://dx.doi.org/10.2139/ssrn.2541477>.
- Pan J, Poteshman AM (2006) The information in option volume for future stock prices. *Rev. Financial Stud.* 19(3):871–908.
- Peng L, Xiong W (2006) Investor attention, overconfidence, and category learning. *J. Financial Econom.* 80(3):563–602.
- Roll R (1984) A simple implicit measure of the effective bid-ask spread in an efficient market. *J. Finance* 39(4):1127–1139.
- Shumway T (2001) Forecasting bankruptcy more accurately: A simple hazard model. *J. Bus.* 74(1):101–124.
- Stilger PS, Kostakis A, Poon S-H (2017) What does risk-neutral skewness tell us about future stock returns? *Management Sci.* 63(6):1814–1834.
- Wang H, Zhou H, Zhou Y (2013) Credit default swap spreads and variance risk premia. *J. Banking Finance* 37(10):3733–3746.
- Xing Y, Zhang X, Zhao R (2010) What does the individual option volatility smirk tell us about future equity returns? *J. Financial Quant. Anal.* 45(3):641–662.
- Zhang BY, Zhou H, Zhu H (2009) Explaining credit default swap spreads with the equity volatility and jump risks of individual firms. *Rev. Financial Stud.* 22(12):5099–5131.